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Probability Theory and Random Variables

1. INTRODUCTION

This chapter offers some background material in probability theory for those readers not already familiar with this subject. The level of presentation corresponds to that in Parzen [6] or Papoulis [5], that is, it is nonmeasure theoretic. Our development is by no means complete. Only those topics are included which are required in the subsequent chapters. Although the presentation is somewhat terse (this is not a text on probability theory), a number of examples are included for the purpose of illustrating various concepts. Readers desiring a more complete and detailed account of the theory are referred to the books by Parzen and Papoulis, which have already been cited. Those readers with a knowledge of measure theory might consult Loeve [4]. For the benefit of those who are familiar with measure theory, we note some of the relationships with that subject in footnotes.

Before proceeding to the subject proper, we delineate and define probability theory and the role it plays in "problem solving," which is, after all, our ultimate objective. We have in mind a physical phenomenon (*experiment*) which is random. That is to say, the outcome of any one occurrence (*trial*) of the phenomenon (*experiment*) cannot be predicted, but is rather governed by chance. For concreteness, consider the rolling

of a fair die. The problem is to determine the number of times an even number will come up in a thousand rolls.

We proceed as follows: (1) There are six equally likely possible outcomes (*events*) on a single roll of a fair die: the six sides. Therefore, the probability of any one side coming up on a single roll is $\frac{1}{6}$. (2) Since the six possible events are mutually exclusive, that is, the occurrence of one excludes the others, and three of the possible events result in the event *an even number comes up*, we conclude that the probability of an even number on a single roll is $\frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{1}{2}$. (3) We therefore predict that an even number will come up about 500 times in 1000 rolls.

It is seen that the solution of the foregoing problem proceeds in three distinct steps. These are

1. determination of (*prior*) probabilities of certain events,
2. operation on these probabilities with certain *rules* to obtain probabilities of other events, and
3. making physical predictions on the basis of the probabilities obtained in Step 2.

Now probability theory deals *only* with Step 2. It consists of certain *rules* derived from *axioms* by deduction. The axioms are, of course, consistent with our intuitive and experimental notions of probabilities. These rules tell us how to operate on the probabilities of certain events to obtain probabilities of other events. Steps 1 and 3 belong to the science of statistics and will not concern us here.

It is worth noting that there are generally three ways of assigning prior probabilities (Step 1). These are

(a) *relative frequency:* conducting a large number of experiments and computing the ratios N_A/N, where N_A is the number of times event A occurs and N is the total number of experiments;

(b) *classical method:* prior probabilities are found *a priori* without experimentation by counting all the equally likely alternatives;

(c) assigning prior probabilities as a *measure of belief*, based on the precept that subjective knowledge is better than no knowledge at all.

In our example of the die experiment, we used method (b) to assign prior probabilities. Method (a) is widely used in practice, especially in experiments that are conceptually too difficult to apply the classical method (b). We shall, in fact, use relative frequency notions to motivate some of our definitions whose rigorous treatment is beyond the scope of this book.

2. PROBABILITY AXIOMS

The underlying concept in probability theory is that of the **probability space**, which we denote by Ω, with **elements** denoted by ω. Ω is the basic sample space, and its elements ω are samples or experimental outcomes. Certain subsets of Ω (collections of outcomes) are called **events**. The class of events will be defined more precisely later.

Referring to the problem discussed in the introductory section, the rolling of a die, we might take $\Omega = \{1, 2, 3, 4, 5, 6\}$, where the element $1 \leqslant \omega \leqslant 6$ is the experimental outcome *the number ω turns up when the die is rolled.* $A_1 = \{2\} \subset \Omega$ is the event *2 turns up when the die is rolled*, and $A_2 = \{2, 4, 6\} \subset \Omega$ is the event *an even number turns up when the die is rolled.* If on a particular trial of this experiment the outcome is $\omega = 2$, then both events A_1 and A_2 occur on that trial. It is clear, that in order to proceed with a probabilistic analysis, all events have to have probabilities assigned to them. In the die experiment, all the ω sets (subsets of Ω) are events and can have assigned probabilities.

Recall the following set theory notation. If A and B are two sets, then $A \cup B$ is their *union*, $A \cap B$ is their *intersection*. $A - B$ is the *complement* of B with respect to A. $\varnothing$ denotes the *empty set*. If $A \cap B = \varnothing$, that is, the sets A and B are *disjoint*, then the events A and B are said to be **mutually exclusive**.

We assign probabilities to events via a **probability function** $\Pr\{\cdot\}$ defined on the class of events. That is, to each event A we assign a number $\Pr\{A\}$, called the *probability of* A. The probability function satisfies the following axioms:

(i) $\Pr\{A\} \geqslant 0$;

(ii) $\Pr\{\Omega\} = 1$;

(iii) if $A_i \cap A_j = \varnothing$, $i \neq j$, $i, j = 1,\ldots, n$, then
$\Pr\{A_1 \cup A_2 \cup \cdots \cup A_n\} = \Pr\{A_1\} + \cdots + \Pr\{A_n\}$;

(iv) if $A_i \cap A_j = \varnothing$, $i \neq j$, $i, j = 1,\ldots, n,\ldots$, then
$\Pr\{A_1 \cup \cdots \cup A_n \cup \cdots\} = \Pr\{A_1\} + \cdots + \Pr\{A_n\} + \cdots$.[1]

Probability theory is based on these axioms. It consists of rules obtained from these axioms by deduction. These axioms, of course, conform with the relative frequency interpretation of probabilities.

[1] As is evident from the axioms, $\Pr\{\cdot\}$ is a completely additive, nonnegative set function that maps ω sets into the positive real numbers, normalized so that $\Pr\{\Omega\} = 1$. The class of events will be a class of measurable sets, and the probability function is then nothing more than a probability measure.

From the probability axioms, it is, for example, easy to show that

$$\Pr\{\varnothing\} = 0, \tag{2.1}$$

$$\Pr\{A\} = 1 - \Pr\{\Omega - A\} \leqslant 1, \tag{2.2}$$

$$\begin{aligned}\Pr\{A_1 \cup A_2\} &= \Pr\{A_1\} + \Pr\{A_2\} - \Pr\{A_1 \cap A_2\} \\ &\leqslant \Pr\{A_1\} + \Pr\{A_2\},\end{aligned} \tag{2.3}$$

$$\text{if } A_1 \subset A_2, \text{ then } \Pr\{A_2\} = \Pr\{A_1\} + \Pr\{A_2 \cap (\Omega - A_1)\} \geqslant \Pr\{A_1\}. \tag{2.4}$$

PROOF: (2.1): Since $\Omega \cap \varnothing = \varnothing$ and $\Omega = \Omega \cup \varnothing$, we have, from Axiom (iii),

$$\Pr\{\Omega\} = \Pr\{\Omega \cup \varnothing\} = \Pr\{\Omega\} + \Pr\{\varnothing\},$$

and then (2.1) follows by Axiom (ii).

(2.2): Since $(\Omega - A) \cap A = \varnothing$ and $(\Omega - A) \cup A = \Omega$, we have from Axiom (iii),

$$\Pr\{\Omega\} = \Pr\{(\Omega - A) \cup A\} = \Pr\{\Omega - A\} + \Pr\{A\}.$$

Now using Axiom (ii),

$$\Pr\{A\} = 1 - \Pr\{\Omega - A\} \leqslant 1,$$

where the last inequality follows from Axiom (i).

(2.3): The sets $A_1 \cup A_2$ and A_2 can be written as unions of two disjoint sets:

$$\begin{aligned}A_1 \cup A_2 &= A_1 \cup [(\Omega - A_1) \cap A_2], \\ A_2 &= (A_1 \cap A_2) \cup [(\Omega - A_1) \cap A_2].\end{aligned}$$

Then by Axiom (iii),

$$\begin{aligned}\Pr\{A_1 \cup A_2\} &= \Pr\{A_1\} + \Pr\{(\Omega - A_1) \cap A_2\}, \\ \Pr\{A_2\} &= \Pr\{A_1 \cap A_2\} + \Pr\{(\Omega - A_1) \cap A_2\}.\end{aligned}$$

Eliminating $\Pr\{(\Omega - A_1) \cap A_2\}$ from these last two equations, we get (2.3). The inequality in (2.3), of course, follows from Axiom (i). (2.4): Since $A_1 \subset A_2$,

$$A_2 = A_1 \cup [(\Omega - A_1) \cap A_2],$$

which is the union of two disjoint sets. Equation (2.4) then follows by Axioms (iii) and (i). ■

We have yet to define the class of events on which the probability function is defined. In the die experiment, we can define $\Pr\{\cdot\}$ on the class of all the subsets of Ω. That is, probabilities can be assigned to all the ω sets; all ω sets are events. This is unfortunately not true in general.[2] Now in defining the class of events, we want the set operations (unions, intersections, complements) performed on events to yield sets that are also events. A class of sets having these properties is called a Borel field. More precisely, a class F of ω sets is called a **Borel field** if

(1) $\Omega \in F$;

(2) if $A \in F$, then $\Omega - A \in F$;

(3) if $A_1, A_2, \ldots, A_n \in F$, then $\bigcup_1^n A_i \in F$ and $\bigcap_1^n A_i \in F$;

(4) if $A_1, A_2, \ldots \in F$, then $\bigcup_1^\infty A_i \in F$ and $\bigcap_1^\infty A_i \in F$.

We state without proof that, given a class F_0 of ω sets, there is a unique Borel field of ω sets $B(F_0)$ with the properties

(1) $F_0 \subset B(F_0)$;

(2) if F_1 is a Borel field of ω sets and if $F_0 \subset F_1$, then $B(F_0) \subset F_1$.

$B(F_0)$ is the *smallest* Borel field of ω sets which contains all the sets of F_0. It is called the **Borel field generated by** F_0. If, for example, $\Omega = \{\textit{real numbers}\}$, we can take $F_0 = \{$intervals of the form $\omega \leqslant x_1$, for all $x_1\}$. Then $B(F_0)$ contains all the open and closed intervals, points, and all countable unions and intersections of intervals and points, that is, all the sets on the real line which are of interest in applications. If $\Omega = E^2$ (the plane), B consists of rectangles and all countable unions and intersections of rectangles. For $\Omega = E^n$ (Euclidean n-space), B is similarly defined.

The **class of events** is a Borel field B and the probability function is defined on B. We suppose that the probability space Ω, Borel field B, and probability function $\Pr\{\cdot\}$ have been defined. The triplet $(\Omega, B, \Pr)$ is called an **experiment**. It will be understood that all sets referred to are Borel sets, that is, sets belonging to B.

Example 2.1. Consider again the experiment of rolling a die (once). We can define a probability space $\Omega_1 = \{1, 2, 3, 4, 5, 6\}$. We can take the set of all subsets of Ω_1 as our Borel field B_1. However, if we are only interested in betting on the events $\{\textit{even}\}$ or $\{\textit{odd}\}$, we can take the

[2] All ω sets may not be measurable. It may be impossible to assign probabilities to all ω sets and satisfy probability Axiom (iv).

field $B_2 = \{\varnothing, \{1, 3, 5\}, \{2, 4, 6\}, \Omega_1\}$. It is easily seen how to define a probability function on B_1 and B_2. Note, however, that

$$A = \{\varnothing, \{1, 3, 5\}, \{2, 4, 6\}, \Omega_1, \{1\}\}$$

is not a field, since $\{1\} \cup \{2, 4, 6\} = \{1, 2, 4, 6\} \notin A$, even though $\{1, 2, 4, 6\} \subset \Omega_1$ and $\Omega_1 \in A$. ▲

Even the definition of a probability space for a given physical experiment is not unique. In our die experiment, we could define Ω_2 as the set of all real numbers. The probability of any Ω_2 subset is then $\frac{1}{6}$ the number of the points 1, 2,..., 6 which it contains. Thus it is clear that the experiment $(\Omega, B, \Pr)$ must be specified. It is not determined by the physical experiment.

We note that $\Pr\{A\} = 0$ does not imply that $A = \varnothing$. For instance, in Ω_2 of Example 2.1, $\Pr\{(7, 8)\} = 0$, but that open interval is obviously not empty. Similarly, $\Pr\{A\} = 1$ does not imply that $A = \Omega$. If P is a property that defines a Borel set and $\Pr\{P\} = 1$, we say that the property P is true **with probability 1** (abbreviated as wp 1). The equivalent terminology **almost everywhere** or **for almost all** ω is also used. $\Pr\{P\} = 1$ means that the probability that property P does not hold is zero.

3. RANDOM VARIABLES

A real finite-valued function $\mathrm{x}(\cdot)$ defined on Ω is called a (real) **random variable** if, for every real number x, the inequality

$$\mathrm{x}(\omega) \leqslant x \tag{2.5}$$

defines an ω set whose probability is defined.[3] This means that the ω set defined by inequality (2.5) is an event. We use lower case roman letters to denote random variables. Lower case italic letters denote **sample values** or **realizations** of the random variable. Thus x is a random variable and x is its realization. That is, for some $\omega \in \Omega$, $\mathrm{x}(\omega) = x$. We often use the abbreviation rv for random variable.

In view of the definition of a random variable, the function

$$F_{\mathrm{x}}(x) \triangleq \Pr\{\mathrm{x}(\omega) \leqslant x\} \tag{2.6}$$

is defined for all real x and is called the **distribution function** of the

[3] A random variable is a Borel measurable function.

random variable x. The subscript x on $F_x(x)$ serves to identify the random variable. In view of (2.4), $F_x(x)$ is monotone, nondecreasing

$$F_x(x_1) \leqslant F_x(x_2) \qquad \text{if} \qquad x_1 < x_2 . \tag{2.7}$$

It is easy to see that

$$\lim_{x \to -\infty} F_x(x) = 0 \tag{2.8}$$

and

$$\lim_{x \to \infty} F_x(x) = 1. \tag{2.9}$$

The distribution function completely describes the properties of an rv. That is, given $F_x(x)$, we can compute probabilities that the rv x takes on values in any set on the real line which is of interest. For example, if $x_1 < x_2$, then

$$\{x(\omega) \leqslant x_2\} = \{x(\omega) \leqslant x_1\} \cup \{x_1 < x(\omega) \leqslant x_2\},$$

where the two ω sets on the right-hand side are disjoint. Therefore, by probability axiom (iii) and the definition of the distribution function, we have

$$\Pr\{x_1 < x(\omega) \leqslant x_2\} = F_x(x_2) - F_x(x_1). \tag{2.10}$$

Now let $x_1 = x - \epsilon$ and $x_2 = x$ in (2.10). Then

$$\Pr\{x - \epsilon < x(\omega) \leqslant x\} = F_x(x) - F_x(x - \epsilon).$$

Letting $\epsilon \to 0$,

$$\Pr\{x(\omega) = x\} = F_x(x) - F_x(x^-). \tag{2.11}$$

So if $F_x(\cdot)$ is discontinuous at x, its jump equals $\Pr\{x(\omega) = x\}$. Now

$$\{x_1 \leqslant x(\omega) \leqslant x_2\} = \{x(\omega) = x_1\} \cup \{x_1 < x(\omega) \leqslant x_2\},$$

so that

$$\begin{aligned} \Pr\{x_1 \leqslant x(\omega) \leqslant x_2\} &= F_x(x_1) - F_x(x_1^-) + F_x(x_2) - F_x(x_1) \\ &= F_x(x_2) - F_x(x_1^-). \end{aligned} \tag{2.12}$$

Similarly,

$$\Pr\{x(\omega) < x_1\} = F_x(x_1^-), \tag{2.13}$$

$$\Pr\{x(\omega) \geqslant x_1\} = 1 - F_x(x_1^-), \tag{2.14}$$

$$\Pr\{x(\omega) > x_1\} = 1 - F_x(x_1), \tag{2.15}$$

and so on. The reader may compute the probabilities of other sets.

A random variable x is called **discrete** if there exists a **mass function** $m_{\mathrm{x}}(\cdot)$ such that

$$F_{\mathrm{x}}(x) = \sum_{\substack{\xi \leqslant x \\ m_{\mathrm{x}}(\xi) > 0}} m_{\mathrm{x}}(\xi). \tag{2.16}$$

It follows from (2.16) and (2.11) that

$$F_{\mathrm{x}}(x) - F_{\mathrm{x}}(x^-) = m_{\mathrm{x}}(x) = \Pr\{\mathrm{x}(\omega) = x\} \tag{2.17}$$

and in view of (2.16) and (2.9)

$$\sum_{\substack{\xi \\ m_{\mathrm{x}}(\xi) > 0}} m_{\mathrm{x}}(\xi) = 1. \tag{2.18}$$

Essentially, a discrete rv can assume only a countable number of values (realizations).

Example 2.2. Consider the experiment of tossing a coin. $\Omega = \{h, t\}$ where h is *heads* and t is *tails*. $B = \{\varnothing, h, t, \Omega\}$. $\Pr\{h\} = p$, $\Pr\{t\} = q$, $p + q = 1$. Define the rv x by $\mathrm{x}(h) = 1$, $\mathrm{x}(t) = 0$. x is a so-called *zero-one* (discrete) random variable associated with the event h. We determine the distribution function $F_{\mathrm{x}}(x)$. If $x \geqslant 1$, then

$$\{\mathrm{x}(\omega) \leqslant x\} = \{\mathrm{x}(\omega) = 0\} \cup \{\mathrm{x}(\omega) = 1\} = \Omega,$$

so that

$$F_{\mathrm{x}}(x) = \Pr\{\mathrm{x}(\omega) \leqslant x\} = 1 \qquad (x \geqslant 1).$$

If $0 \leqslant x < 1$, then

$$\{\mathrm{x}(\omega) \leqslant x\} = \{\mathrm{x}(\omega) = 0\} = t,$$

so that

$$F_{\mathrm{x}}(x) = q \qquad (0 \leqslant x < 1).$$

If $x < 0$, then $\{\mathrm{x}(\omega) \leqslant x\} = \varnothing$, so that $F_{\mathrm{x}}(x) = 0$. Combining these results we have

$$F_{\mathrm{x}}(x) = \begin{cases} 1, & x \geqslant 1 \\ q, & 0 \leqslant x < 1 \\ 0, & x < 0. \end{cases}$$

It is easy to see that there exists a mass function that is given by

$$m_{\mathrm{x}}(x) = \begin{cases} q, & x = 0 \\ p, & x = 1 \\ 0, & \text{otherwise.} \end{cases}$$

The distribution and mass functions are displayed in Figs. 2.1 and 2.2, respectively. ▲

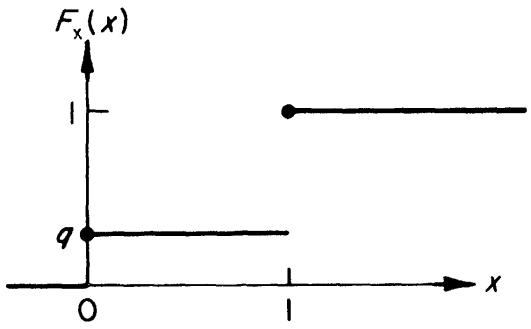


FIG 2.1. Distribution function of a discrete random variable.

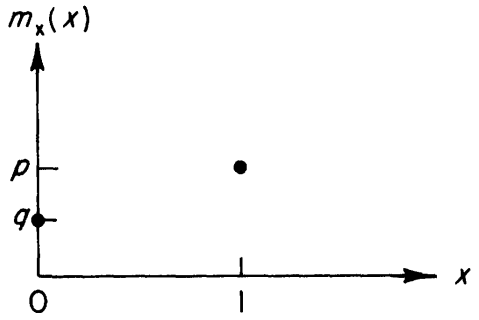


FIG. 2.2. Probability mass function.

A random variable x is called **continuous** if there exists a **density function** $p_x(\cdot)$ such that

$$F_x(x) = \int_{-\infty}^{x} p_x(\xi)\, d\xi, \qquad -\infty < x < \infty.^4 \tag{2.19}$$

The density function exists if the distribution function is absolutely continuous, that is, the number of points at which $F_x(\cdot)$ is not differentiable is countable. It is immediately evident from (2.19) that $F_x(x)$ is then continuous at all x, and that

$$p_x(x) = \frac{d}{dx} F_x(x) \tag{2.20}$$

at all x at which the derivative exists.

Since $F_x(\cdot)$ is continuous, it follows from (2.11) that

$$\Pr\{x(\omega) = x\} = 0. \tag{2.21}$$

Since $F_x(\cdot)$ is monotonic, and in view of (2.19),

$$p_x(x) \geqslant 0. \tag{2.22}$$

Also, in view of (2.19) and (2.9),

$$\int_{-\infty}^{\infty} p_x(\xi)\, d\xi = 1. \tag{2.23}$$

[4] In general, we assume that this integral and, until further notice, all other integrals in this book are *Riemann* integrals. This will lead us into some subtle mathematical difficulties that need not, however, concern an engineer. We shall explain these difficulties as they arise.

A function $p_{\mathrm{x}}(\cdot)$ satisfying (2.22) and (2.23) is a density function. That is, it defines a distribution function which in turn determines an experiment $(\Omega, B, \Pr)$ and a random variable (see Papoulis [5, p. 99]).

Now in view of (2.19), for $x_1 < x_2$,

$$F_{\mathrm{x}}(x_2) - F_{\mathrm{x}}(x_1) = \int_{x_1}^{x_2} p_{\mathrm{x}}(\xi)\, d\xi.$$

Therefore, using (2.10),

$$\Pr\{x_1 \leqslant \mathrm{x}(\omega) \leqslant x_2\} = \int_{x_1}^{x_2} p_{\mathrm{x}}(\xi)\, d\xi,$$

and, for Δx sufficiently small,

$$\Pr\{x \leqslant \mathrm{x}(\omega) \leqslant x + \Delta x\} \simeq p_{\mathrm{x}}(x)\, \Delta x. \tag{2.24}$$

Thus we have a complete analogy between a probability density and a mass density, if we imagine the mass distributed continuously along the real line.

Example 2.3. A telephone call occurs at random in the time interval $[0, T]$. Let $\Omega = [0, T]$; $\Pr\{t_1 \leqslant \omega \leqslant t_2\} = (t_2 - t_1)/T$; $t_1, t_2 \in [0, T]$. Define the rv x by $\mathrm{x}(\omega) = \omega$. We determine its distribution and density functions. If $x > T$, then

$$\{\mathrm{x}(\omega) \leqslant x\} = \Omega, \qquad F_{\mathrm{x}}(x) = 1.$$

If $0 \leqslant x \leqslant T$, then

$$\{\mathrm{x}(\omega) \leqslant x\} = \{0 \leqslant \omega \leqslant x\}, \qquad F_{\mathrm{x}}(x) = x/T.$$

If $x < 0$, then

$$\{\mathrm{x}(\omega) \leqslant x\} = \varnothing, \qquad F_{\mathrm{x}}(x) = 0.$$

Differentiating this distribution function, we get

$$p_{\mathrm{x}}(x) = \begin{cases} 0, & x < 0 \\ 1/T, & 0 < x < T \\ 0, & x > T. \end{cases}$$

The distribution function is not differentiable at 0 and T, but it is easy to see that the density function may be arbitrarily defined at these two points without affecting the experiment. These distribution and density functions are displayed in Figs. 2.3 and 2.4. ▲

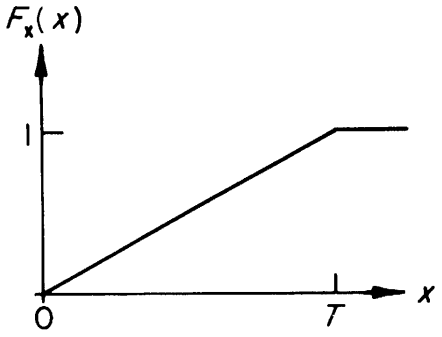


FIG. 2.3. Uniform distribution function.

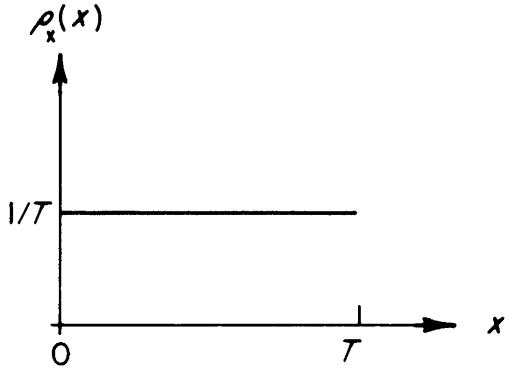


FIG. 2.4. Uniform density function.

The random variable of this example is called a **uniformly distributed** (continuous) random variable (uniformly distributed on the interval $[0, T]$). Its density function could have been written down immediately from the probability function by observing that every interval of equal length in $[0, T]$ is equally probable, and intervals outside $[0, T]$ are impossible. In engineering applications, it is common to characterize an rv by directly specifying its density function.

Consider the "random variable" $\mathrm{x}(\omega) = c$ for all $\omega \in \Omega$. This rv is the constant c, independent of the experimental outcomes. For $x < c$,

$$\Pr\{\mathrm{x}(\omega) \leqslant x\} = F_{\mathrm{x}}(x) = 0,$$

and, for $x \geqslant c$,

$$\Pr\{\mathrm{x}(\omega) \leqslant x\} = F_{\mathrm{x}}(x) = 1.$$

Thus,

$$F_{\mathrm{x}}(x) = H(x - c), \tag{2.25}$$

where $H(\cdot)$ is the Heaviside unit step function. Proceeding formally, we have

$$p_{\mathrm{x}}(x) = \frac{d}{dx} F_{\mathrm{x}}(x) = \delta(x - c), \tag{2.26}$$

where $\delta(\cdot)$ is the Dirac delta function. Thus we see that constants may be formally regarded as random variables, with density functions that are Dirac delta functions.

In view of the fact that the density (mass) function determines the distribution function, the density (mass) function completely characterizes a continuous (discrete) random variable. In this book, we will be concerned only with continuous random variables. We henceforth assume that all our continuous rv's possess density functions with all the required smoothness properties. Thus, instead of specifying an experiment $(\Omega, B, \Pr)$ and defining the rv x on Ω, we shall equivalently

specify a random variable with a given probability density function. Properties of discrete rv's can be written by analogy with continuous rv's by replacing the density function by the mass function and integrals by sums. We note, in passing, that there are random variables that are of the so-called mixed type. That is, their distribution function cannot be written as in (2.16) or in (2.19), but rather as a combination of integrals and sums. We shall not encounter such rv's in this book.

The **expectation** (*mean, average, first moment*) of a (continuous) random variable x is defined (when it exists) by

$$\mathscr{E}\{\mathrm{x}\} \triangleq \int_{-\infty}^{\infty} x p_{\mathrm{x}}(x)\, dx.^{5} \tag{2.27}$$

We say that $\mathscr{E}\{\mathrm{x}\}$ *exists* if $\mathscr{E}\{|\,\mathrm{x}\,|\} < \infty$, that is, if the integral in (2.27) is absolutely convergent. Therefore, $\mathscr{E}\{\mathrm{x}\}$ exists if and only if $\mathscr{E}\{|\,\mathrm{x}\,|\} < \infty$, and then

$$|\,\mathscr{E}\{\mathrm{x}\}| \leqslant \mathscr{E}\{|\,\mathrm{x}\,|\}. \tag{2.28}$$

In writing the integral in (2.27) we shall generally, in the future, omit the limits of integration. It will be understood that, in the absence of limits, the integral is over the whole real line.

A *fixed* (nonrandom) real function $f(\cdot)$ of the random variable x is also a random variable,[6] say $\mathrm{y} = f(\mathrm{x})$. Its expectation may be computed in two ways:

$$\mathscr{E}\{\mathrm{y}\} \triangleq \int y p_{\mathrm{y}}(y)\, dy = \int f(x)\, p_{\mathrm{x}}(x)\, dx.^{7} \tag{2.29}$$

Let a and b be fixed constants and $\mathrm{y} = a$. Then

$$\mathscr{E}\{a\} = a \tag{2.30}$$

in view of (2.23). Now let $\mathrm{y} = ag_1(\mathrm{x}) + bg_2(\mathrm{x})$. Then

$$\mathscr{E}\{ag_1(\mathrm{x}) + bg_2(\mathrm{x})\} = a\mathscr{E}\{g_1(\mathrm{x})\} + b\mathscr{E}\{g_2(\mathrm{x})\}. \tag{2.31}$$

[5] The expectation can also be written in terms of the *Stieltjes* integral $\int_{-\infty}^{\infty} x\, dF_{\mathrm{x}}(x)$. A definition of the expectation in terms of this latter integral is more general than ours. In view of (2.20) and our assumption of the existence and smoothness of the density function, it is equivalent to our definition.

[6] Under the mild condition that $\{x : f(x) \leqslant y\} \in B$ for every y. This ensures that $\{\mathrm{y}[\mathrm{x}(\omega)] \leqslant y\} \in B$ (is an event). This means that $f(\cdot)$ is a *Borel* measurable (*Baire*) function. It is assumed that all functions in this book are Baire functions.

[7] For the last integral to exist as a Riemann integral, $f(\cdot)$ must be more restricted, e.g., Riemann integrable over every interval. That the two evaluations of $\mathscr{E}\{\mathrm{y}\}$ are equal follows from the properties of the integral. The proof, however, is beyond the scope of this book.

Now

$$\mathscr{E}\{\mathrm{x}^n\} \triangleq \int x^n p_{\mathrm{x}}(x)\, dx \tag{2.32}$$

defines the ***n*th moment** of x. $\mathscr{E}\{\mathrm{x}^2\}$ is called the **mean square** value of x.

$$\mathscr{E}\{(\mathrm{x} - \mathscr{E}\{\mathrm{x}\})^n\} \triangleq \int (x - \mathscr{E}\{\mathrm{x}\})^n p_{\mathrm{x}}(x)\, dx \tag{2.33}$$

defines the ***n*th moment of** x **about the mean (*n*th central moment).** The second moment about the mean is called the **variance** of x:

$$\mathrm{var}\{\mathrm{x}\} = \mathscr{E}\{(\mathrm{x} - \mathscr{E}\{\mathrm{x}\})^2\} = \mathscr{E}\{\mathrm{x}^2\} - \mathscr{E}^2\{\mathrm{x}\}. \tag{2.34}$$

Thus the variance is the mean square minus the square mean. Finally, the **standard deviation** of x is defined by

$$\sigma\{\mathrm{x}\} \triangleq (\mathrm{var}\{\mathrm{x}\})^{1/2}. \tag{2.35}$$

It is a measure of the *dispersion* about the mean in the samples of x. Note the analogy between the variance and the mass moment of inertia about the center of mass.

Example 2.4. Referring to the random variable of Example 2.3, it is easy to compute the mean and variance to be $m = \mathscr{E}\{\mathrm{x}\} = T/2$, $\sigma^2 = T^2/12$. ▲

Example 2.5. It was noted that the variance of a random variable is a measure of the dispersion in its samples. We show here that, if c is a fixed constant and if $\mathscr{E}\{(x - c)^2\} = 0$, then $\mathrm{x} = c$ wp 1. Suppose the assertion is not true. Then there exists an $\epsilon > 0$ such that

$$\Pr\{|\,\mathrm{x} - c\,| > \epsilon\} = \int_{|x-c|>\epsilon} p_{\mathrm{x}}(x)\, dx > 0.$$

But then

$$\mathscr{E}\{(\mathrm{x} - c)^2\} = \int_{-\infty}^{\infty} (x - c)^2 p_{\mathrm{x}}(x)\, dx \geqslant \int_{|x-c|>\epsilon} (x - c)^2 p_{\mathrm{x}}(x)\, dx$$

$$\geqslant \epsilon^2 \int_{|x-c|>\epsilon} p_{\mathrm{x}}(x)\, dx > 0.$$

In particular, if $\mathscr{E}\{\mathrm{x}^2\} = 0$, then $\mathrm{x} = 0$ wp 1. ▲

A random variable may also be specified in terms of its **characteristic function** defined by

$$\varphi_{\mathrm{x}}(u) \triangleq \mathscr{E}\{\exp(iu\mathrm{x})\}, \qquad i^2 = -1. \tag{2.36}$$

The importance of this function lies in the fact that one can obtain all the moments of the rv x by differentiating its characteristic function. It is easy to show that

$$\mathscr{E}\{\mathrm{x}^n\} = (1/i^n)\frac{d^n}{du^n}\varphi_{\mathrm{x}}(0) \tag{2.37}$$

for all n, provided of course that $\varphi_{\mathrm{x}}(u)$ is analytic. The characteristic function always exists since

$$|\mathscr{E}\{\exp(iu\mathrm{x})\}| \leqslant \mathscr{E}\{|\exp(iu\mathrm{x})|\} = 1.$$

From its definition, we see that the characteristic function is just the Fourier transform of the density function

$$\varphi_{\mathrm{x}}(u) = \int_{-\infty}^{\infty} \exp(iux)\, p_{\mathrm{x}}(x)\, dx. \tag{2.38}$$

Therefore, if the characteristic function is absolutely integrable, the inverse Fourier transform

$$p_{\mathrm{x}}(x) = (1/2\pi)\int_{-\infty}^{\infty} \exp(-iux)\, \varphi_{\mathrm{x}}(u)\, du \tag{2.39}$$

gives the density function.

Of central importance in this book will be random variables that are **Gaussian**, or equivalently, **normally distributed**. A random variable is Gaussian if its density function is given by

$$p_{\mathrm{x}}(x) = (1/2\pi\sigma^2)^{1/2} \exp\left[-\frac{1}{2}\left(\frac{x-m}{\sigma}\right)^2\right], \tag{2.40}$$

where m and σ^2 are (constant) parameters. An easy computation shows that

$$m = \mathscr{E}\{\mathrm{x}\}, \qquad \sigma^2 = \mathrm{var}\{\mathrm{x}\}. \tag{2.41}$$

Note that these two parameters characterize the Gaussian density. If the rv x is Gaussian with mean m and variance σ^2, we sometimes write

$$\mathrm{x} \sim N(m, \sigma^2). \tag{2.42}$$

The normal density function has the familiar bell shape. The normal distribution function is, of course, given by

$$F_x(x) = (1/2\pi\sigma^2)^{1/2} \int_{-\infty}^{x} \exp\left[-\frac{1}{2}\left(\frac{\xi - m}{\sigma}\right)^2\right] d\xi. \tag{2.43}$$

The importance of Gaussian random variables will become apparent in subsequent chapters. Readers interested in other density functions that are important in applications may consult Papoulis [5, Chapter 4], for example. The normal density and distribution functions (with $m = 0$ and $\sigma^2 = 1$) are plotted in Figs. 2.5 and 2.6.

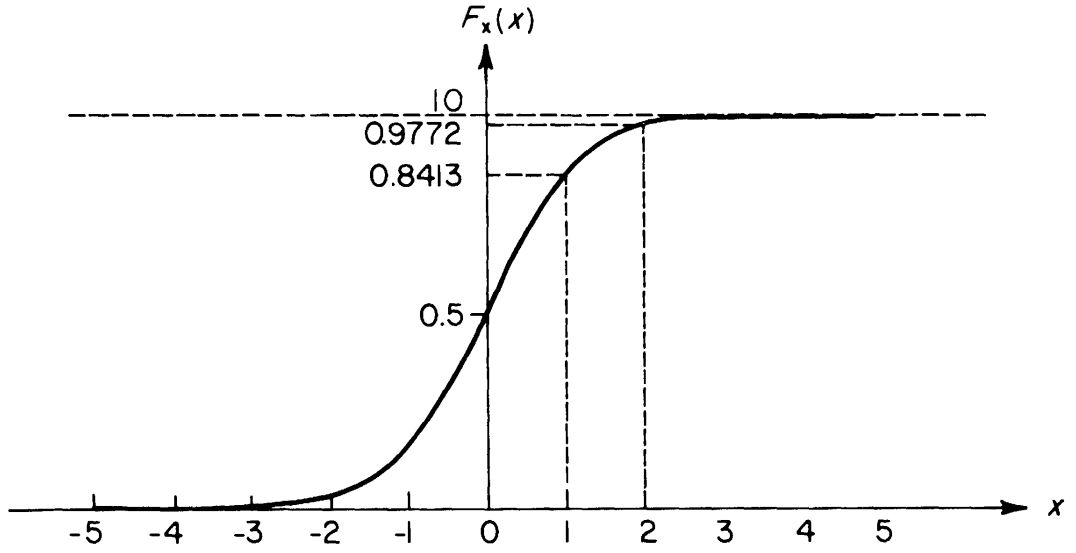


FIG. 2.5. Normal distribution function.

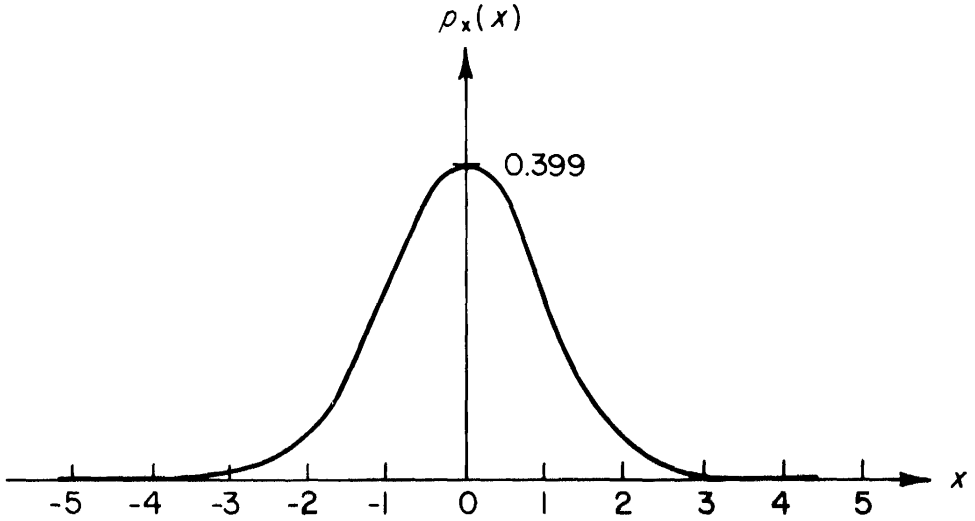


FIG. 2.6. Normal density function.

Example 2.6. Let the rv x be normally distributed with mean m and variance σ^2. Compute its characteristic function

$$\varphi_x(u) = (1/2\pi\sigma^2)^{1/2} \int_{-\infty}^{\infty} \exp\left[iux - \frac{1}{2}\left(\frac{x-m}{\sigma}\right)^2\right] dx.$$

Letting $y = (x - m)/\sigma$,

$$\begin{aligned}\varphi_x(u) &= (1/2\pi)^{1/2} \exp(ium) \int_{-\infty}^{\infty} \exp(iu\sigma y - \tfrac{1}{2}y^2)\, dy \\ &= (1/2\pi)^{1/2} \exp(ium - \tfrac{1}{2}u^2\sigma^2) \int_{-\infty}^{\infty} \exp[-\tfrac{1}{2}(y - iu\sigma)^2]\, dy \\ &= \exp(ium - \tfrac{1}{2}u^2\sigma^2). \end{aligned} \tag{2.44}$$

Let us compute all the central moments of x (moments about the mean). In view of the symmetry about the mean of the Gaussian density function (2.40), it is evident that all the odd central moments are zero. Since the central moments of x are the same as the moments of $x - m$ (from definition), we compute the characteristic function of $x - m$ and use (2.37). From the definition of the characteristic function (2.36) and (2.44),

$$\begin{aligned}\varphi_{x-m}(u) &= \exp(-ium)\, \varphi_x(u) \\ &= \exp(-\tfrac{1}{2}u^2\sigma^2). \end{aligned} \tag{2.45}$$

We may note from (2.45) that the central moments of x will depend only on its variance σ^2. It is easy to establish by induction that

$$\varphi^{(n+2)}(u) = -(n+1)\,\sigma^2\varphi^{(n)}(u) - u\sigma^2\varphi^{(n+1)}(u), \qquad n = 0, 1, \ldots, \tag{2.46}$$

with $\varphi^{(0)}(u) = \varphi(u)$, $\varphi^{(1)}(u) = -u\sigma^2\varphi(u)$. We have omitted the subscript $x - m$ on φ. The superscript denotes the derivative

$$\varphi^{(n)}(u) = (d^n/du^n)\, \varphi(u).$$

Evaluating (2.46) at $u = 0$,

$$\varphi^{(n+2)}(0) = -(n+1)\,\sigma^2\varphi^{(n)}(0). \tag{2.47}$$

Consider (2.47) as a difference equation with initial condition $\varphi^{(1)}(0) = 0$. It immediately follows that

$$\varphi^{(n)}(0) = 0, \qquad \text{all odd} \quad n \geqslant 1.$$

With initial condition $\varphi^{(0)}(0) = 1$, (2.47) has the solution

$$\varphi^{(n)}(0) = (i^n)\, 1 \cdot 3 \cdot 5 \cdots (n-1)\,\sigma^n, \qquad \text{all even} \quad n \geqslant 2.$$

Therefore, in view of (2.37),

$$\mathscr{E}\{(\mathrm{x} - m)^n\} = \begin{cases} 0, & \text{all odd} \quad n \geqslant 1 \\ 1 \cdot 3 \cdot 5 \cdots (n-1)\,\sigma^n, & \text{all even} \quad n \geqslant 2. \end{cases} \tag{2.48}$$

We see that for any $\sigma > 0$, no matter how small, the even central moments grow without bound ($n \to \infty$). ▲

4. JOINTLY DISTRIBUTED RANDOM VARIABLES

The (continuous) random variables $\mathrm{x}_1, \mathrm{x}_2, \ldots, \mathrm{x}_n$ are said to be **jointly distributed** if they are defined on the same probability space. They may be characterized by their **joint distribution function**

$$F_{\mathrm{x}_1,\ldots,\mathrm{x}_n}(x_1, \ldots, x_n) \triangleq \Pr\{\mathrm{x}_1(\omega) \leqslant x_1, \ldots, \mathrm{x}_n(\omega) \leqslant x_n\}, \tag{2.49}$$

where

$$\{\mathrm{x}_1(\omega) \leqslant x_1, \ldots, \mathrm{x}_n(\omega) \leqslant x_n\} = \{\mathrm{x}_1(\omega) \leqslant x_1\} \cap \cdots \cap \{\mathrm{x}_n(\omega) \leqslant x_n\},$$

or by their **joint density function**

$$F_{\mathrm{x}_1,\ldots,\mathrm{x}_n}(x_1, \ldots, x_n) = \int_{-\infty}^{x_1} \cdots \int_{-\infty}^{x_n} p_{\mathrm{x}_1,\ldots,\mathrm{x}_n}(\xi_1, \ldots, \xi_n)\, d\xi_1, \ldots, d\xi_n\,. \tag{2.50}$$

It follows from (2.50) that

$$p_{\mathrm{x}_1,\ldots,\mathrm{x}_n}(x_1, \ldots, x_n) = \frac{\partial^n}{\partial x_1, \ldots, \partial x_n} F_{\mathrm{x}_1,\ldots,\mathrm{x}_n}(x_1, \ldots, x_n) \tag{2.51}$$

at all $(x_1, \ldots, x_n)$ at which the foregoing derivative exists. The properties of the joint distribution and density functions follow by analogy with the single variable case of the preceding section and will not be developed here.

Example 2.7. Let x and y be jointly distributed rv's with joint distribution function $F_{\mathrm{x},\mathrm{y}}(x, y)$ and joint density $p_{\mathrm{x},\mathrm{y}}(x, y)$. Convince yourself from the definitions and reasoning analogous to that used in Eqs. (2.7)–(2.15) and (2.21)–(2.24) that

$$F_{\mathrm{x},\mathrm{y}}(-\infty, y) = F_{\mathrm{x},\mathrm{y}}(x, -\infty) = F_{\mathrm{x},\mathrm{y}}(-\infty, -\infty) = 0,$$

$$F_{\mathrm{x},\mathrm{y}}(\infty, \infty) = 1,$$

$$F_{\mathrm{x},\mathrm{y}}(x_2, y) - F_{\mathrm{x},\mathrm{y}}(x_1, y) = \Pr\{x_1 < \mathrm{x}(\omega) \leqslant x_2, \mathrm{y}(\omega) \leqslant y\} \geqslant 0,$$

$(x_2 > x_1)$, and similarly in y. Also,

$$p_{\mathrm{x},\mathrm{y}}(x, y) \geqslant 0,$$

$$\Pr\{x < \mathrm{x}(\omega) \leqslant x + \Delta x, y < \mathrm{y}(\omega) \leqslant y + \Delta y\} \simeq p_{\mathrm{x},\mathrm{y}}(x, y)\, \Delta x\, \Delta y,$$

$$\int\int p_{\mathrm{x},\mathrm{y}}(x, y)\, dx\, dy = 1. \quad \blacktriangle$$

We are often interested only in *some* of the rv's $\mathrm{x}_1, \ldots, \mathrm{x}_n$; in $\mathrm{x}_1, \ldots, \mathrm{x}_m$, $m < n$, say. Their joint distribution function is

$$F_{\mathrm{x}_1,\ldots,\mathrm{x}_m}(x_1, \ldots, x_m) = F_{\mathrm{x}_1,\ldots,\mathrm{x}_n}(x_1, \ldots, x_m, \infty, \ldots, \infty), \tag{2.52}$$

as is evident from (2.49). The distribution in (2.52) is sometimes called **marginal**. Differentiating (2.52) and using (2.50) and (2.51), we obtain the **marginal density function**

$$p_{\mathrm{x}_1,\ldots,\mathrm{x}_m}(x_1, \ldots, x_m) = \int_{-\infty}^{\infty} \cdots \int p_{\mathrm{x}_1,\ldots,\mathrm{x}_n}(x_1, \ldots, x_n)\, dx_{m+1} \cdots dx_n. \tag{2.53}$$

In view of this and the definition (2.27), the **expectation** of x_k, $1 \leqslant k \leqslant n$, is given by

$$\mathscr{E}\{\mathrm{x}_k\} = \int \cdots \int x_k p_{\mathrm{x}_1,\ldots,\mathrm{x}_n}(x_1, \ldots, x_n)\, dx_1, \ldots, dx_n. \tag{2.54}$$

Example 2.8. Let x and y be jointly **uniformly** distributed on $[0, T] \times [0, T]$, that is,

$$p_{\mathrm{x},\mathrm{y}}(x, y) = \begin{cases} 1/T^2 & \text{on } [0, T] \times [0, T] \\ 0 & \text{elsewhere.} \end{cases}$$

The marginal densities are

$$p_{\mathrm{x}}(x) = \int_{-\infty}^{\infty} p_{\mathrm{x},\mathrm{y}}(x, y)\, dy = \int_0^T (1/T^2)\, dy = 1/T \qquad (x \in [0, T]),$$

$$p_{\mathrm{x}}(x) = 0 \qquad (x \notin [0, T]),$$

and, similarly,

$$p_{\mathrm{y}}(y) = \begin{cases} 1/T, & y \in [0, T] \\ 0, & \text{elsewhere.} \end{cases}$$

Therefore,

$$p_{\mathrm{x},\mathrm{y}}(x, y) = p_{\mathrm{x}}(x)\, p_{\mathrm{y}}(y);$$

x and y are *independent.* Compute $\Pr\{x \leqslant T/2, y \leqslant T/2\} = 1/4$. Compute $\mathscr{E}\{\mathrm{x}\} = \mathscr{E}\{\mathrm{y}\} = T/2$. ▲

In general, if f is a fixed, real function of the jointly distributed random variables $\mathrm{x}_1, \ldots, \mathrm{x}_n$, say $\mathrm{y} = f(\mathrm{x}_1, \ldots, \mathrm{x}_n)$, then

$$\mathscr{E}\{\mathrm{y}\} \triangleq \int y p_{\mathrm{y}}(y)\, dy = \int \cdots \int f(x_1, \ldots, x_n)\, p_{\mathrm{x}_1,\ldots,\mathrm{x}_n}(x_1, \ldots, x_n)\, dx_1 \cdots dx_n \quad (2.55)$$

when it exists. The expectation of x_k has already been given in (2.54). The **second moment** of x_k is

$$\mathscr{E}\{\mathrm{x}_k^2\} = \int \cdots \int x_k^2 p_{\mathrm{x}_1,\ldots,\mathrm{x}_n}(x_1, \ldots, x_n)\, dx_1 \cdots dx_n . \quad (2.56)$$

Higher moments and central moments are written in an obvious way. In addition, joint moments of the form

$$\mathscr{E}\{\mathrm{x}_k^{\alpha} \mathrm{x}_l^{\beta}\}$$

and joint central moments

$$\mathscr{E}\{(\mathrm{x}_k - \mathscr{E}\{\mathrm{x}_k\})^{\alpha} (\mathrm{x}_l - \mathscr{E}\{\mathrm{x}_l\})^{\beta}\}$$

can be defined. Of particular importance is the **covariance** of x_k and x_l which is defined by

$$\operatorname{cov}\{\mathrm{x}_k, \mathrm{x}_l\} \triangleq \mathscr{E}\{(\mathrm{x}_k - \mathscr{E}\{\mathrm{x}_k\})(\mathrm{x}_l - \mathscr{E}\{\mathrm{x}_l\})\}. \quad (2.57)$$

We have that

$$\operatorname{cov}\{\mathrm{x}_k, \mathrm{x}_l\} = \mathscr{E}\{\mathrm{x}_k \mathrm{x}_l\} - \mathscr{E}\{\mathrm{x}_k\}\, \mathscr{E}\{\mathrm{x}_l\}. \quad (2.58)$$

Note that

$$\operatorname{cov}\{\mathrm{x}_k, \mathrm{x}_k\} = \operatorname{var}\{\mathrm{x}_k\}.$$

The ratio

$$\frac{\operatorname{cov}\{\mathrm{x}_k, \mathrm{x}_l\}}{\sigma\{\mathrm{x}_k\}\, \sigma\{\mathrm{x}_l\}} = \rho(\mathrm{x}_k, \mathrm{x}_l) \quad (2.59)$$

defines the **correlation coefficient** of x_k and x_l. It is, of course, assumed in (2.59) that x_k and x_l are nondegenerate (have positive variances) and that their variances are finite. See Example 2.5.

Having defined various moments and joint moments, it is important to know under what conditions these moments exist. We now develop several useful inequalities and use these to establish conditions for the existence of moments.

Let x and y be jointly distributed rv's. We first note that the expectation operator $\mathscr{E}$ is *linear*. That is, if a and b are fixed constants, then

$$\mathscr{E}\{ax + by\} = a\mathscr{E}\{x\} + b\mathscr{E}\{y\}. \tag{2.60}$$

This fact was already used [in (2.58)] and, of course, follows from like properties of the integral that defines the expectation. See also (2.31).

Now consider

$$\mathscr{E}\{(a \mid x \mid - \mid y \mid)^2\} = a^2\mathscr{E}\{\mid x \mid^2\} - 2a\mathscr{E}\{\mid xy \mid\} + \mathscr{E}\{\mid y \mid^2\} \geqslant 0$$

for all a. This is a quadratic in a which is nonnegative for all a. As a result, it cannot have two real, distinct roots, so its discriminant cannot be positive; that is,

$$\mathscr{E}^2\{\mid xy \mid\} - \mathscr{E}\{\mid x \mid^2\}\mathscr{E}\{\mid y \mid^2\} \leqslant 0.$$

This is just *Schwarz's* inequality ($\mathscr{E}$ operator is an integral operator)

$$\mathscr{E}^2\{\mid xy \mid\} \leqslant \mathscr{E}\{\mid x \mid^2\}\mathscr{E}\{\mid y \mid^2\}. \tag{2.61}$$

The inequality

$$\mathscr{E}\{\mid x + y \mid\} \leqslant \mathscr{E}\{\mid x \mid\} + \mathscr{E}\{\mid y \mid\} \tag{2.62}$$

follows from the triangle inequality for real numbers (just take expectations of both sides). Using the triangle inequality, it is also easy to verify that

$$\mathscr{E}\{\mid x + y \mid^2\} \leqslant 2\mathscr{E}\{\mid x \mid^2\} + 2\mathscr{E}\{\mid y \mid^2\}. \tag{2.63}$$

Now we can easily prove the following theorems.

Theorem 2.1. *If* $\mathscr{E}\{\mid x \mid^2\} < \infty$, *then* $\mathscr{E}\{\mid x \mid\} < \infty$.

PROOF: Taking the square root in Schwarz's inequality and setting $y \equiv 1$, we have

$$\mathscr{E}\{\mid x \mid\} \leqslant \mathscr{E}^{1/2}\{\mid x \mid^2\} < \infty. \quad \blacksquare$$

Thus the existence of the second moment implies the existence of the first moment. In fact, the following theorem can be proved (see Loeve [4, p. 155]).

Theorem 2.2. *If* $\mathscr{E}\{\mid x \mid^n\} < \infty$, *then* $\mathscr{E}\{\mid x \mid^m\} < \infty$ *for all* $1 \leqslant m \leqslant n$.

The existence of the nth moment guarantees the existence of all lower-order moments.

Theorem 2.3. *If* $\mathscr{E}\{|\,x\,|^2\} < \infty$ *and* $\mathscr{E}\{|\,y\,|^2\} < \infty$, *then* (i) $\mathscr{E}\{|\,x + y\,|^2\} < \infty$ *and* (ii) $\mathscr{E}\{|\,xy\,|\} < \infty$.

PROOF: (i) follows with the aid of (2.63), and (ii) via Schwarz's inequality. ■

The following theorem can also be proved (see Loeve [4, p. 155]).

Theorem 2.4. *If* $\mathscr{E}\{|\,x\,|^n\} < \infty$ *and* $\mathscr{E}\{|\,y\,|^n\} < \infty$, *then* $\mathscr{E}\{|\,x + y\,|^n\} < \infty$.

Example 2.9. Let x and y be random variables with *finite second moments*. That is, $\mathscr{E}\{|\,x\,|^2\} < \infty$ and $\mathscr{E}\{|\,y\,|^2\} < \infty$. By Theorem 2.1, $\mathscr{E}\{x\}$ and $\mathscr{E}\{y\}$ exist. Therefore, var{x} and var{y} exist. By Theorem 2.3, $\mathscr{E}\{xy\}$ exists; therefore cov{x, y} also exists. ▲

Random variables with finite second moments constitute an important class of rv's. We shall have much more to say about them in Chapter 3.

Two jointly distributed random variables x_1 and x_2 are said to be **independent** if any of the following *equivalent* conditions is satisfied:

$$F_{x_1,x_2}(x_1, x_2) = F_{x_1}(x_1)F_{x_2}(x_2), \qquad \text{all } x_1, x_2, \tag{2.64}$$

$$p_{x_1,x_2}(x_1, x_2) = p_{x_1}(x_1)p_{x_2}(x_2), \qquad \text{all } x_1, x_2, \tag{2.65}$$

$$\mathscr{E}\{f(x_1)g(x_2)\} = \mathscr{E}\{f(x_1)\}\mathscr{E}\{g(x_2)\}, \tag{2.66}$$

for all fixed functions $f(\cdot)$, $g(\cdot)$, provided these expectations exist. This concept generalizes to more than two random variables. We say that $x_1, \ldots, x_n$ are **mutually independent** if

$$p_{x_1,\ldots,x_n}(x_1, \ldots, x_n) = p_{x_1}(x_1) \cdots p_{x_n}(x_n) \tag{2.67}$$

for all $x_1, \ldots, x_n$. Conditions (2.64) and (2.66) generalize in an obvious way. Also, $x_1, \ldots, x_n$ are said to be **jointly independent** of $y_1, \ldots, y_m$ if

$$p_{x_1,\ldots,x_n,y_1,\ldots,y_m}(x_1, \ldots, x_n, y_1, \ldots, y_m) = p_{x_1,\ldots,x_n}(x_1, \ldots, x_n)\, p_{y_1,\ldots,y_m}(y_1, \ldots, y_m) \tag{2.68}$$

for all $x_1, \ldots, x_n, y_1, \ldots, y_m$.

We say that the two jointly distributed random variables x_k and x_l are **uncorrelated** if their second moments are finite and if

$$\text{cov}\{x_k, x_l\} = 0. \tag{2.69}$$

Their correlation coefficient (2.59) is then zero.

Theorem 2.5. *Let the jointly distributed random variables* x_k *and* x_l *be independent. Then they are also uncorrelated.*

PROOF:

$$\begin{aligned}\operatorname{cov}\{\mathrm{x}_k, \mathrm{x}_l\} &= \iint x_k x_l p_{\mathrm{x}_k,\mathrm{x}_l}(x_k, x_l)\, dx_k\, dx_l - \mathscr{E}\{\mathrm{x}_k\}\,\mathscr{E}\{\mathrm{x}_l\} \\ &= \iint x_k x_l p_{\mathrm{x}_k}(x_k)\, p_{\mathrm{x}_l}(x_l)\, dx_k\, dx_l - \mathscr{E}\{\mathrm{x}_k\}\,\mathscr{E}\{\mathrm{x}_l\} \\ &= \int x_k p_{\mathrm{x}_k}(x_k)\, dx_k \cdot \int x_l p_{\mathrm{x}_l}(x_l)\, dx_l - \mathscr{E}\{\mathrm{x}_k\}\,\mathscr{E}\{\mathrm{x}_l\} \\ &= \mathscr{E}\{\mathrm{x}_k\}\cdot\mathscr{E}\{\mathrm{x}_l\} - \mathscr{E}\{\mathrm{x}_k\}\,\mathscr{E}\{\mathrm{x}_l\} = 0. \quad \blacksquare\end{aligned}$$

Note that the converse to Theorem 2.5 is, in general, not true (see Theorem 2.6, however). Lack of correlation means that (2.66) holds for functions $f(\mathrm{x}) = g(\mathrm{x}) = \mathrm{x}$, whereas independence requires (2.66) to hold for *all* functions f and g.

Example 2.10. Consider the two random variables y_1 and y_2 defined by

$$\mathrm{y}_1 = \sin 2\pi\mathrm{x}, \qquad \mathrm{y}_2 = \cos 2\pi\mathrm{x},$$

where x is uniformly distributed on [0, 1]. It is easy to compute $\mathscr{E}\{\mathrm{y}_1\} = \mathscr{E}\{\mathrm{y}_2\} = \mathscr{E}\{\mathrm{y}_1\mathrm{y}_2\} = 0$. Therefore, $\operatorname{cov}\{\mathrm{y}_1, \mathrm{y}_2\} = 0$, so that y_1 and y_2 are uncorrelated. However, $\mathrm{y}_1^2 + \mathrm{y}_2^2 = 1$, so y_1 and y_2 are clearly not independent. ▲

Two jointly distributed rv's x_k and x_l are said to be **orthogonal** if their second moments are finite and if

$$\mathscr{E}\{\mathrm{x}_k\mathrm{x}_l\} = 0. \tag{2.70}$$

Note that if x_k and x_l are uncorrelated, then $\mathrm{x}_k - \mathscr{E}\{\mathrm{x}_k\}$ and $\mathrm{x}_l - \mathscr{E}\{\mathrm{x}_l\}$ are orthogonal. Also, if x_k and x_l are uncorrelated and either $\mathscr{E}\{\mathrm{x}_k\}$ or $\mathscr{E}\{\mathrm{x}_l\}$ is zero, then they are orthogonal.

Example 2.11. Let x and y be uncorrelated. Then the variance of their sum is the sum of their variances. ▲

Example 2.12. Let x and y be rv's with joint density function

$$p_{\mathrm{x},\mathrm{y}}(x, y) = (1/2\pi)\exp[-(1/2)(x^2 + y^2)].$$

Let the reader show that x and y are independent, uncorrelated, orthogonal and are both *identically* distributed (they are both Gaussian). See (2.40). ▲

Example 2.13. We note here the properties of the correlation coefficient (2.59). If, in the derivation of Schwarz's inequality (2.61), we replace $|\,x\,|$ by $x - \mathscr{E}\{x\}$ and $|\,y\,|$ by $y - \mathscr{E}\{y\}$, we obtain

$$\mathscr{E}^2\{(x - \mathscr{E}\{x\})(y - \mathscr{E}\{y\})\} \leqslant \mathscr{E}\{(x - \mathscr{E}\{x\})^2\}\,\mathscr{E}\{(y - \mathscr{E}\{y\})^2\}.$$

Thus $\operatorname{cov}^2\{x, y\} \leqslant \operatorname{var}\{x\} \operatorname{var}\{y\}$, and we immediately have that $0 \leqslant |\,\rho(x, y)| \leqslant 1$. Obviously $\rho(x, y) = 0$ if and only if x and y are uncorrelated. We now show that $|\,\rho\,| = 1$ if and only if $y = ax + b$, where a and b are fixed constants.[8] Suppose $y = ax + b$. Then

$$\mathscr{E}\{y\} = a\mathscr{E}\{x\} + b,$$

$$\operatorname{var}\{y\} = a^2 \operatorname{var}\{x\},$$

$$\operatorname{cov}\{x, y\} = a \operatorname{var}\{x\},$$

so that $\rho = \pm 1$. Now suppose $|\,\rho\,| = 1$. Then $\operatorname{cov}^2\{x, y\} = \operatorname{var}\{x\} \operatorname{var}\{y\}$. Let

$$a = \frac{\operatorname{cov}\{x, y\}}{\operatorname{var}\{x\}},$$

$$b = \mathscr{E}\{y\} - \frac{\operatorname{cov}\{x, y\}\,\mathscr{E}\{x\}}{\operatorname{var}\{x\}},$$

and use the result in Example 2.5. ▲

The **characteristic function** of the jointly distributed rv's $x_1, \ldots, x_n$ is defined by

$$\varphi_{x_1,\ldots,x_n}(u_1, \ldots, u_n) \triangleq \mathscr{E}\left\{\exp\left(i \sum_{j=1}^{n} u_j x_j\right)\right\}. \tag{2.71}$$

The **marginal** characteristic function of $x_1, \ldots, x_m$, $m < n$, is clearly

$$\varphi_{x_1,\ldots,x_n}(u_1, \ldots, u_m, 0, \ldots, 0). \tag{2.72}$$

If $x_1, \ldots, x_n$ are *independent*, then [see (2.66) or (2.67)]

$$\varphi_{x_1,\ldots,x_n}(u_1, \ldots, u_n) = \varphi_{x_1}(u_1) \cdots \varphi_{x_n}(u_n). \tag{2.73}$$

[8] We can only show that $y = ax + b$ wp 1. We often do not make a distinction between identities and wp 1 equalities. For engineering purposes, they are the same.

The converse can also be shown to be true. Moments and joint moments of the rv's $x_1, \ldots, x_n$ can be obtained from their joint characteristic function by analogy with the single variable case.

The random variables x_1 and x_2 are said to be **Gaussian** or **jointly normally distributed** if their joint density function is given by

$$p_{x_1,x_2}(x_1, x_2) = \frac{1}{2\pi\sigma\{x_1\}\,\sigma\{x_2\}(1-\rho^2)^{1/2}} \exp\left\{-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x_1 - \mathscr{E}\{x_1\}}{\sigma\{x_1\}}\right)^2 - 2\rho\left(\frac{x_1 - \mathscr{E}\{x_1\}}{\sigma\{x_1\}}\right)\left(\frac{x_2 - \mathscr{E}\{x_2\}}{\sigma\{x_2\}}\right) + \left(\frac{x_2 - \mathscr{E}\{x_2\}}{\sigma\{x_2\}}\right)^2\right]\right\}, \quad (2.74)$$

where $\rho(x_1, x_2)$ is the correlation coefficient. We see that the joint Gaussian density in (2.74) is undefined if $|\rho| = 1$. See Example 2.13. In that case, however, x_1 completely determines x_2 and *vice versa*, and it is sufficient to specify the density of just one of these rv's. Also compare (2.74) with Example 2.12. Now x_2 may be integrated out in (2.74) to show that the marginal density of x_1 is Gaussian with mean $\mathscr{E}\{x_1\}$ and variance $\sigma^2\{x_1\}$. See also Theorem 2.12. We may note that if x_1 and x_2 are uncorrelated ($\rho = 0$), then (2.74) reduces to

$$p_{x_1,x_2}(x_1, x_2) = p_{x_1}(x_1)\, p_{x_2}(x_2), \quad (2.75)$$

where the two densities on the right-hand side are Gaussian [see (2.40)]. Thus, two Gaussian rv's that are uncorrelated are also independent. Compare this with Theorem 2.5, the comments following it, and Example 2.10.

In dealing with jointly distributed random variables, it will be very convenient to arrange these rv's in a vector and take advantage of the tools of vector-matrix algebra. Let $x_1, \ldots, x_n$ be jointly distributed rv's. We introduce the **random vector** x:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \qquad \mathbf{x}^T = [x_1, \ldots, x_n]. \quad (2.76)$$

A vector will always be a column vector. Superscript T denotes the vector (matrix) transpose. Thus $\mathbf{x}^T$ is a row vector. We make no notational distinction between a random vector x and a (scalar) random variable x. This is because many of our equations can stand equally well with x interpreted as a scalar or as a vector. For example,

$$p_x(x)$$

can be interpreted as the density function of the scalar rv x or as the joint density of $x_1, \ldots, x_n$ which we could write in longhand (if we cared to) as

$$p_{x_1,\ldots,x_n}(x_1, \ldots, x_n).$$

Also,

$$\int p_x(x)\, dx = 1$$

could mean (for a scalar rv x) the integral over the whole real line of the density of the scalar rv x. If we interpret dx as the volume element in E^n (n-dimensional Euclidean space), then the preceding integral can be interpreted as the integral over E^n:

$$\int \cdots \int p_{x_1,\ldots,x_n}(x_1, \ldots, x_n)\, dx_1 \cdots dx_n = 1.$$

The reason why a notational distinction is not important is, of course, the fact that joint density functions are scalar functions. In any event, when we write x we shall either state whether it is a scalar or a vector, if it does matter, or else that fact will be obvious from the context.

If x is a random vector, its **expectation** is the vector

$$\mathcal{E}\{\mathbf{x}\} = \begin{bmatrix} \mathcal{E}\{x_1\} \\ \vdots \\ \mathcal{E}\{x_n\} \end{bmatrix}, \qquad \mathcal{E}\{\mathbf{x}^T\} = [\mathcal{E}\{x_1\}, \ldots, \mathcal{E}\{x_n\}]. \tag{2.77}$$

$\mathcal{E}\{\mathbf{x}\}$ is often called the **mean vector** (of x). If $A = [a_{ij}]$ is a **random matrix** (matrix whose elements are random variables), then, similarly,

$$\mathcal{E}\{A\} = [\mathcal{E}\{a_{ij}\}]. \tag{2.78}$$

The square ($n \times n$) matrix of elements $\text{cov}\{x_i, x_j\}$ is called the **covariance matrix** of the random vector x and is often designated by P_x:

$$P_x = [\text{cov}\{x_i, x_j\}] = \mathcal{E}\{(\mathbf{x} - \mathcal{E}\{\mathbf{x}\})(\mathbf{x} - \mathcal{E}\{\mathbf{x}\})^T\}$$

$$= \begin{bmatrix} \text{var}\{x_1\} & \text{cov}\{x_1, x_2\} & \cdots & \text{cov}\{x_1, x_n\} \\ \text{cov}\{x_1, x_2\} & \text{var}\{x_2\} & \cdots & \text{cov}\{x_2, x_n\} \\ \vdots & \vdots & \vdots & \vdots \\ \text{cov}\{x_1, x_n\} & \text{cov}\{x_2, x_n\} & \cdots & \text{var}\{x_n\} \end{bmatrix}. \tag{2.79}$$

The random matrix $(\mathbf{x} - \mathcal{E}\{\mathbf{x}\})(\mathbf{x} - \mathcal{E}\{\mathbf{x}\})^T$ is, we point out, the outer product of the random vector $(\mathbf{x} - \mathcal{E}\{\mathbf{x}\})$ with itself. We note that the

covariance matrix is *symmetric.* It can be shown that it is *positive semidefinite.*[9] This follows immediately, for the case $n = 2$, from the inequality in Example 2.13. Other inequalities can be similarly developed to prove this fact for arbitrary n. Note that, in case the elements of x are (pairwise) uncorrelated, the covariance matrix P_{x} is *diagonal.* We note that higher-order joint moments (of x_i, x_j) cannot be expressed in matrix notation. They would be three-dimensional arrays (tensors).

Using the vector-matrix notation just developed, we write the (joint) density function of n **jointly normally distributed** random variables $\mathrm{x}_1, \ldots, \mathrm{x}_n$:

$$p_{\mathrm{x}}(x) = [(2\pi)^{n/2} \mid P_{\mathrm{x}} \mid^{1/2}]^{-1} \exp[-\tfrac{1}{2}(x - \mathscr{E}\{\mathrm{x}\})^{\mathrm{T}} P_{\mathrm{x}}^{-1}(x - \mathscr{E}\{\mathrm{x}\})]. \quad (2.80)$$

$|\cdot|$ denotes the determinant, $|P_{\mathrm{x}}| = \det P_{\mathrm{x}}$, and superscript -1 denotes the matrix inverse. Compare this with (2.74) for $n = 2$. Note that the parameters $\mathscr{E}\{\mathrm{x}\}$ and P_{x} completely characterize the multivariate Gaussian density function. If the random vector x is Gaussian with mean vector $\mathscr{E}\{\mathrm{x}\}$ and covariance matrix P_{x}, we sometimes write

$$\mathrm{x} \sim N(\mathscr{E}\{\mathrm{x}\}, P_{\mathrm{x}}). \quad (2.81)$$

We note that the Gaussian density is defined only if the covariance matrix P_{x} is positive definite. Analogously to the $n = 2$ case and Example 2.13, if P_{x} is only semidefinite, say of rank $(n - 1)$, there exists a wp 1 linear relationship between the x_i. One of these rv's can therefore be eliminated.

We have already noted that, if the elements of x are uncorrelated, then P is diagonal. But then, as it is easy to see, Eq. (2.80) reduces to

$$p_{\mathrm{x}}(x) = p_{\mathrm{x}_1}(x_1) \cdots p_{\mathrm{x}_n}(x_n). \quad (2.82)$$

We therefore have (compare with Theorem 2.5):

Theorem 2.6. *If the jointly normally distributed random variables* $\mathrm{x}_1, \ldots, \mathrm{x}_n$ *are (pairwise) uncorrelated, then they are independent.*

Example 2.14. Using vector-matrix notation, we rewrite the joint characteristic function in (2.71) as

$$\varphi_{\mathrm{x}}(u) = \mathscr{E}\{\exp(iu^{\mathrm{T}}\mathrm{x})\}, \quad (2.83)$$

[9] The matrix A is said to be *positive definite (positive semidefinite)* iff $x^{\mathrm{T}}Ax > 0$ ($x^{\mathrm{T}}Ax \geq 0$) for all vectors $x \neq 0$. If A is positive definite (positive semidefinite), we often write $A > 0$ ($A \geq 0$).

where

$$\mathrm{x}^{\mathrm{T}} = [\mathrm{x}_1, \ldots, \mathrm{x}_n], \qquad u^{\mathrm{T}} = [u_1, \ldots, u_n].$$

It is not very difficult to verify that if $\mathrm{x}_1, \ldots, \mathrm{x}_n$ are jointly *normally* distributed, then

$$\varphi_{\mathrm{x}}(u) = \exp(iu^{\mathrm{T}}\mathscr{E}\{\mathrm{x}\} - \tfrac{1}{2}u^{\mathrm{T}}P_{\mathrm{x}}u), \tag{2.84}$$

where P_{x} is the covariance matrix of (2.79). Note that the characteristic function exists even if P_{x} is semidefinite. ▲

Let the random n-vector y be a (vector-valued) function of the random n-vector x, say $\mathrm{y} = f(\mathrm{x})$. If we know the (joint) density function $p_{\mathrm{x}}(x)$, we can determine $p_{\mathrm{y}}(y)$ in terms of it.

Theorem 2.7. *Let* x, y *be random n-vectors with* $\mathrm{y} = f(\mathrm{x})$. *Suppose* f^{-1} *exists and both* f *and* f^{-1} *are continuously differentiable. Then*

$$p_{\mathrm{y}}(y) = p_{\mathrm{x}}[f^{-1}(y)] \left\| \frac{\partial f^{-1}(y)}{\partial y} \right\|, \tag{2.85}$$

where $\| \partial f^{-1}(y)/\partial y \| > 0$ *is the absolute value of the Jacobian determinant.*

PROOF: For an arbitrary set $S \subset E^n$,

$$\Pr\{\mathrm{y}(\omega) \in S\} = \Pr\{f[\mathrm{x}(\omega)] \in S\} = \Pr\{\mathrm{x}(\omega) \in f^{-1}(S)\}.$$

But

$$\Pr\{\mathrm{y}(\omega) \in S\} = \int_S p_{\mathrm{y}}(y)\, dy$$

and

$$\begin{aligned} \Pr\{\mathrm{x}(\omega) \in f^{-1}(S)\} &= \int_{f^{-1}(S)} p_{\mathrm{x}}(x)\, dx \\ &= \int_S p_{\mathrm{x}}[f^{-1}(y)] \left\| \frac{\partial f^{-1}(y)}{\partial y} \right\| dy, \end{aligned}$$

where the last integral follows by well-known rules for transformation of integrals. Therefore,

$$\int_S \left[p_{\mathrm{y}}(y) - p_{\mathrm{x}}[f^{-1}(y)] \left\| \frac{\partial f^{-1}(y)}{\partial y} \right\| \right] dy = 0 \tag{2.86}$$

for arbitrary S. Suppose (2.85) is not true. Then there is some $\bar{y} \in E^n$ at which the integrand in (2.86) is positive (say). But then, by (assumed)

continuity, there is some region S containing $\bar{y}$ in which the integrand in (2.86) is positive. This contradicts (2.86). ■

The density $p_y(y)$ in (2.85) is sometimes called a **derived density,** since it is derived from the density function of x. The assumptions we made on the function f in Theorem 2.7 can be relaxed to obtain more general relationships for the derived density. See Papoulis [5, pp. 126, 201] for more general results in the cases of $n = 1$ and $n = 2$. Numerous special cases are also treated in Papoulis [5, Chapters 5 and 7]. Theorem 2.7 will have important applications in our work in filtering theory.

Example 2.15. Let $y_1 = x_1 + x_2$. We wish to determine the density of y_1 from the densities of x_1 and x_2. Now this transformation is a projection, and Theorem 2.7 is not directly applicable. Let us adjoin to this the transformation $y_2 = x_2$ (say) and consider

$$y_1 = x_1 + x_2, \qquad y_2 = x_2.$$

In vector-matrix notation,

$$y = Ax,$$

where

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \qquad A^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}, \qquad x^T = [x_1, x_2], \qquad y^T = [y_1, y_2].$$

Now Theorem 2.7 is applicable, and we have

$$p_{y_1,y_2}(y_1, y_2) = p_{x_1,x_2}(y_1 - y_2, y_2). \tag{2.87}$$

We get the (marginal) density of y_1 by integrating y_2 out in (2.87); that is,

$$p_{y_1}(y_1) = \int p_{x_1,x_2}(y_1 - y_2, y_2)\, dy_2. \tag{2.88}$$

In case x_1 and x_2 are *independent*, Eq. (2.88) reduces to

$$p_{y_1}(y_1) = \int p_{x_1}(y_1 - y_2)\, p_{x_2}(y_2)\, dy_2, \tag{2.89}$$

which is a convolution of the densities of x_1 and x_2. If x_1 and x_2 are independent,

$$\begin{aligned} \mathcal{E}\{\exp[iu(x_1 + x_2)]\} &= \mathcal{E}\{\exp(iux_1) \cdot \exp(iux_2)\} \\ &= \mathcal{E}\{\exp(iux_1)\}\, \mathcal{E}\{\exp(iux_2)\}, \end{aligned}$$

so that

$$\varphi_{X_1+X_2}(u) = \varphi_{X_1}(u)\,\varphi_{X_2}(u), \tag{2.90}$$

which is another reason for the importance of characteristic functions. ▲

5. CONDITIONAL PROBABILITIES AND EXPECTATIONS

A rigorous treatment of this subject unfortunately requires rather advanced mathematical tools, and, as a result, we must limit ourselves to a formal treatment. We shall attempt to motivate our definitions via the relative frequency notion of probability. A mathematical treatment of this subject may be found in Loeve [4, Chapter VII] or Doob [3, Chapter I]. Papoulis [5, 2.3, 4.3, 6.2] presents numerous examples and certain cases not treated here.

Given two events A and B, we define the **conditional probability function** $\Pr\{A \mid B\}$ of event A *given* the event B by

$$\Pr\{A \mid B\} \triangleq \frac{\Pr\{A \cap B\}}{\Pr\{B\}} \qquad (\Pr\{B\} > 0). \tag{2.91}$$

$\Pr\{A \mid B\}$ is undefined if $\Pr\{B\} = 0$. Loosely speaking, $\Pr\{A \mid B\}$ is the probability of event A, having already observed the occurrence of event B. Thus it is clearly unimportant that $\Pr\{A \mid B\}$ is undefined if $\Pr\{B\} = 0$.

Example 2.16. Consider the die experiment (Example 2.1) with $\Omega = \{1, 2, 3, 4, 5, 6\}$, Borel field F the class of all subsets of Ω, $\Pr\{i\} = 1/6$, $1 \leqslant i \leqslant 6$. Let the event $B = \{2, 4, 6\} = \{even\}$. Having observed the event *even*, we must revise our probability function. Clearly now $\Pr\{1 \mid B\} = \Pr\{3 \mid B\} = \Pr\{5 \mid B\} = 0$, $\Pr\{2 \mid B\} = \Pr\{4 \mid B\} = \Pr\{6 \mid B\} = \frac{1}{3}$. ▲

From the experiment $(\Omega, F, \Pr\{\cdot\})$, we can construct a new experiment $(\Omega, F, \Pr\{\cdot \mid B\})$. It is easy to show that $\Pr\{\cdot \mid B\}$, as defined in (2.91), satisfies the probability axioms of Section 2.

Our definition of the conditional probability function coincides with the intuitive relative frequency notion of probability. For example, consider an experiment repeated N times, N very large. Let N_A denote the number of occurrences of event A, N_B the number of occurrences of

event B, and N_{AB} the number of occurrences of both A and B. Then, clearly,

$$\Pr\{A\} = N_A/N, \qquad \Pr\{B\} = N_B/N,$$
$$\Pr\{A \cap B\} = N_{AB}/N, \qquad \Pr\{A \mid B\} = N_{AB}/N_B,$$

so that

$$\Pr\{A \mid B\} = N_{AB}/N \cdot N/N_B = \Pr\{A \cap B\}/\Pr\{B\}$$

which coincides with our definition. Now in case A and B are independent [see (2.64)], $\Pr\{A \cap B\} = \Pr\{A\}\Pr\{B\}$, so that

$$\Pr\{A \mid B\} = \Pr\{A\}, \tag{2.92}$$

which adds intuitive content to our definition of independence.

Let x and y be jointly distributed continuous random variables. Given the joint distribution and density of x and y, we seek the distribution and density of x *given* a realization y of y. We shall specify the events A and B in (2.91) in terms of conditions on our rv's, and then proceed formally to define the desired conditional distribution and density functions by a limiting process. Let $A = \{\mathrm{x}(\omega) \leqslant x\}$. From (2.91),

$$\Pr\{\mathrm{x}(\omega) \leqslant x \mid B\} = \frac{\Pr\{\mathrm{x}(\omega) \leqslant x, B\}}{\Pr\{B\}}. \tag{2.93}$$

This is just the distribution function defined on the experiment $(\Omega, F, \Pr\{\cdot \mid B\})$ and, as can easily be shown, has properties (2.7), (2.8), and (2.9). Thus

$$F_{\mathrm{x}\mid B}(x \mid B) = \frac{\Pr\{\mathrm{x}(\omega) \leqslant x, B\}}{\Pr\{B\}}. \tag{2.94}$$

Now let $B = \{y < \mathrm{y}(\omega) \leqslant y + \Delta y\}$ and assume that $p_{\mathrm{y}}(y) > 0$ for the y in question. Then [see (2.10)]

$$\Pr\{B\} = F_{\mathrm{y}}(y + \Delta y) - F_{\mathrm{y}}(y)$$

and (2.94) becomes

$$F_{\mathrm{x}\mid\{y<\mathrm{y}(\omega)\leqslant y+\Delta y\}}(x \mid y < \mathrm{y}(\omega) \leqslant y + \Delta y\}$$
$$= \frac{\Pr\{\mathrm{x}(\omega) \leqslant x, y < \mathrm{y}(\omega) \leqslant y + \Delta y\}}{F_{\mathrm{y}}(y + \Delta y) - F_{\mathrm{y}}(y)}. \tag{2.95}$$

Now the numerator in (2.95) can be written as

$$F_{\mathrm{x},\mathrm{y}}(x, y + \Delta y) - F_{\mathrm{x},\mathrm{y}}(x, y)$$

(see Example 2.7). Dividing the numerator and denominator in (2.95) by Δy, that equation now becomes

$$F_{\mathrm{x}\mid\{y<\mathrm{y}(\omega)\leqslant y+\Delta y\}}(x \mid y < \mathrm{y}(\omega) \leqslant y + \Delta y) = \frac{[F_{\mathrm{x},\mathrm{y}}(x, y + \Delta y) - F_{\mathrm{x},\mathrm{y}}(x, y)]/\Delta y}{[F_{\mathrm{y}}(y + \Delta y) - F_{\mathrm{y}}(y)]/\Delta y}. \tag{2.96}$$

We formally define $F_{\mathrm{x}\mid\mathrm{y}}(x \mid y)$, the distribution x *given* $\{\mathrm{y}(\omega) = y\}$, as the $\Delta y = 0$ limit of (2.96), which is

$$F_{\mathrm{x}\mid\mathrm{y}}(x \mid y) = \frac{\partial F_{\mathrm{x},\mathrm{y}}(x, y)/\partial y}{dF_{\mathrm{y}}(y)/dy} = \frac{\int_{-\infty}^{x} p_{\mathrm{x},\mathrm{y}}(\xi, y)\, d\xi}{p_{\mathrm{y}}(y)}. \tag{2.97}$$

The last equality follows from (2.50) (differentiate). Note also that $p_{\mathrm{y}}(y)$ is the marginal density

$$p_{\mathrm{y}}(y) = \int p_{\mathrm{x},\mathrm{y}}(x, y)\, dx. \tag{2.98}$$

Differentiating (2.97) with respect to x, we get

$$p_{\mathrm{x}\mid\mathrm{y}}(x \mid y) = \frac{p_{\mathrm{x},\mathrm{y}}(x, y)}{p_{\mathrm{y}}(y)}. \tag{2.99}$$

Note the analogy with (2.91).

Motivated by the preceding formal derivation, we *define* the **conditional density function** $p_{\mathrm{x}\mid\mathrm{y}}(x \mid y)$ of x *given* $\{\mathrm{y}(\omega) = y\}$ for all x and y such that $p_{\mathrm{y}}(y) > 0$ by

$$p_{\mathrm{x}\mid\mathrm{y}}(x \mid y) \triangleq \frac{p_{\mathrm{x},\mathrm{y}}(x, y)}{p_{\mathrm{y}}(y)} = \frac{p_{\mathrm{x},\mathrm{y}}(x, y)}{\int p_{\mathrm{x},\mathrm{y}}(x, y)\, dx}.^{10} \tag{2.100}$$

The qualification that $p_{\mathrm{y}}(y) > 0$ is no restriction, since, for any observed realization y of y, $p_{\mathrm{y}}(y) > 0$ wp 1. Reversing the roles of x and y in (2.100),

$$p_{\mathrm{x},\mathrm{y}}(x, y) = p_{\mathrm{y}\mid\mathrm{x}}(y \mid x)\, p_{\mathrm{x}}(x), \tag{2.101}$$

[10] We note in passing that, if the rv's x and y are jointly discrete, then the conditional probability mass function is defined by

$$m_{\mathrm{x}\mid\mathrm{y}}(x \mid y) = \frac{m_{\mathrm{x},\mathrm{y}}(x, y)}{m_{\mathrm{y}}(y)},$$

and the conditional expectation is then

$$\mathscr{E}\{\mathrm{x} \mid \mathrm{y}\} = \sum_{\mathrm{x}} x m_{\mathrm{x}\mid\mathrm{y}}(x \mid y).$$

and, substituting this in (2.100), we have

$$p_{\mathrm{x}|\mathrm{y}}(x \mid y) = \frac{p_{\mathrm{y}|\mathrm{x}}(y \mid x)\, p_{\mathrm{x}}(x)}{p_{\mathrm{y}}(y)}. \tag{2.102}$$

This is known as **Bayes' rule** or **Bayes' theorem**. Now our formal derivation of (2.99) needs only slight modifications in case x and y are jointly distributed random vectors. *Therefore, in the definition* (2.100), *as well as in* (2.101) *and* (2.102), x *and* y *are, in general, random vectors.* We write (2.100) out in "longhand" in the case where x is a random n-vector and y is a random m-vector:

$$\begin{aligned} &p_{\mathrm{x}_1,\ldots,\mathrm{x}_n|\mathrm{y}_1,\ldots,\mathrm{y}_m}(x_1,\ldots,x_n \mid y_1,\ldots,y_m) \\ &\quad = \frac{p_{\mathrm{x}_1,\ldots,\mathrm{x}_n,\mathrm{y}_1,\ldots,\mathrm{y}_m}(x_1,\ldots,x_n,y_1,\ldots,y_m)}{p_{\mathrm{y}_1,\ldots,\mathrm{y}_m}(y_1,\ldots,y_m)} \\ &\quad = \frac{p_{\mathrm{x}_1,\ldots,\mathrm{x}_n,\mathrm{y}_1,\ldots,\mathrm{y}_m}(x_1,\ldots,x_n,y_1,\ldots,y_m)}{\int\cdots\int p_{\mathrm{x}_1,\ldots,\mathrm{x}_n,\mathrm{y}_1,\ldots,\mathrm{y}_m}(\xi_1,\ldots,\xi_n,y_1,\ldots,y_m)\, d\xi_1\cdots d\xi_n}. \end{aligned} \tag{2.100}$$

The **conditional expectation** of the random variable (vector) x *given* the random variable (vector) y (given $\{\mathrm{y}(\omega) = y\}$) is defined [by analogy with (2.27)] by

$$\mathscr{E}\{\mathrm{x} \mid \mathrm{y}\} \triangleq \int x p_{\mathrm{x}|\mathrm{y}}(x \mid y)\, dx. \tag{2.103}$$

If x is a random n-vector, then

$$\mathscr{E}\{\mathrm{x} \mid \mathrm{y}\} = [\mathscr{E}\{\mathrm{x}_1 \mid \mathrm{y}\},\ldots,\mathscr{E}\{\mathrm{x}_n \mid \mathrm{y}\}]^{\mathrm{T}}. \tag{2.104}$$

Compare this with (2.77) and (2.54). Note that, in view of (2.100), $\mathscr{E}\{\mathrm{x} \mid \mathrm{y}\}$ exists if $\mathscr{E}\{\mathrm{x}\}$ exists.

We note that $p_{\mathrm{x}|\mathrm{y}}(x \mid y)$ depends on the realizations of y and, as a result, is *itself* a random variable. It is a function of the random variable (vector) y [see (2.107) below]. Similarly, $\mathscr{E}\{\mathrm{x} \mid \mathrm{y}\}$ is also a random variable (vector), being a function of y [see (2.110) below].

The formal properties of conditional density functions and conditional expectations are summarized in the following theorems.

Theorem 2.8 (Conditional Density Function). *Let* x *and* y *be jointly distributed random variables* (*vectors*). *Then*

$$p_{\mathrm{x}|\mathrm{y}}(x \mid y) \geqslant 0, \tag{2.105}$$

$$\int p_{\mathrm{x}|\mathrm{y}}(x \mid y)\, dx = 1, \tag{2.106}$$

$$p_{\mathrm{x}}(x) = \mathscr{E}\{p_{\mathrm{x}|\mathrm{y}}(x \mid y)\} = \int p_{\mathrm{x}|\mathrm{y}}(x \mid y)\, p_{\mathrm{y}}(y)\, dy, \tag{2.107}$$

$$p_{\mathrm{x}|\mathrm{y}}(x \mid y) = p_{\mathrm{x}}(x) \qquad \text{if} \quad \mathrm{x} \text{ and } \mathrm{y} \text{ are independent.} \tag{2.108}$$

These properties follow immediately from (2.100) and (2.68). The proof is left to the reader.

Theorem 2.9 (Conditional Expectations). *Let* x, y, z *be jointly distributed random variables (vectors); c, d fixed constants; $g(\cdot)$ a scalar-valued function. Assume that* $\mathscr{E}\{\mathrm{x}\}$, $\mathscr{E}\{\mathrm{z}\}$, $\mathscr{E}\{g(\mathrm{y})\,\mathrm{x}\}$ *exist. Then*

$$\mathscr{E}\{\mathrm{x} \mid \mathrm{y}\} = \mathscr{E}\{\mathrm{x}\} \qquad \text{if} \quad \mathrm{x} \text{ and } \mathrm{y} \text{ are independent,} \tag{2.109}$$

$$\mathscr{E}\{\mathrm{x}\} = \mathscr{E}\{\mathscr{E}\{\mathrm{x} \mid \mathrm{y}\}\}, \tag{2.110}$$

$$\mathscr{E}\{g(\mathrm{y})\,\mathrm{x} \mid \mathrm{y}\} = g(\mathrm{y})\,\mathscr{E}\{\mathrm{x} \mid \mathrm{y}\}, \tag{2.111}$$

$$\mathscr{E}\{g(\mathrm{y})\,\mathrm{x}\} = \mathscr{E}\{g(\mathrm{y})\,\mathscr{E}\{\mathrm{x} \mid \mathrm{y}\}\}, \tag{2.112}$$

$$\mathscr{E}\{c \mid \mathrm{y}\} = c, \tag{2.113}$$

$$\mathscr{E}\{g(\mathrm{y}) \mid \mathrm{y}\} = g(\mathrm{y}), \tag{2.114}$$

$$\mathscr{E}\{c\mathrm{x} + d\mathrm{z} \mid \mathrm{y}\} = c\mathscr{E}\{\mathrm{x} \mid \mathrm{y}\} + d\mathscr{E}\{\mathrm{z} \mid \mathrm{y}\}. \tag{2.115}$$

PROOF: These properties follow easily from (2.100), (2.103), and Theorem 2.8. We point out that (2.112) follows from (2.111) with the help of (2.110). As for (2.115),

$$\mathscr{E}\{c\mathrm{x} + d\mathrm{z} \mid \mathrm{y}\} = \int (c\mathrm{x} + d\mathrm{z})\, p_{\mathrm{x},\mathrm{z}|\mathrm{y}}(x, z \mid y)\, dx\, dz,$$

from whence (2.115) follows. The rest of the proof is left to the reader. ■

We comment that (2.110) means, roughly speaking, that to compute the average of x we may compute its average for each y and then average over the y's. In view of (2.115), the conditional expectation operator $\mathscr{E}\{\cdot \mid \mathrm{y}\}$ is linear. It should be obvious that Schwarz's inequality (2.61), as well as inequalities (2.62) and (2.63), hold for conditional expectations. As a result, so do Theorems 2.1 and 2.3.

Example 2.17. The conditional mean (vector) was noted in (2.104). The *conditional covariance matrix* is appropriately [compare with (2.79)] defined by

$$P_{\mathrm{x}|\mathrm{y}} \triangleq \mathscr{E}\{(\mathrm{x} - \mathscr{E}\{\mathrm{x} \mid \mathrm{y}\})(\mathrm{x} - \mathscr{E}\{\mathrm{x} \mid \mathrm{y}\})^{\mathrm{T}} \mid \mathrm{y}\}. \tag{2.116}$$

Clearly, $P_{x|y}$ is a *random matrix* (matrix whose elements are rv's). What about $\mathscr{E}\{P_{x|y}\}$? We now show that

$$P_x = \mathscr{E}\{P_{x|y}\} + P_{\mathscr{E}\{x|y\}}, \tag{2.117}$$

where P_x is the unconditional covariance matrix defined in (2.79) and

$$P_{\mathscr{E}\{x|y\}} = \mathscr{E}\{(\mathscr{E}\{x \mid y\} - \mathscr{E}\{x\})(\mathscr{E}\{x \mid y\} - \mathscr{E}\{x\})^T\} \tag{2.118}$$

is the covariance matrix of the conditional mean.

Using (2.110), we first write

$$P_x = \mathscr{E}\{\mathscr{E}\{(x - \mathscr{E}\{x\})(x - \mathscr{E}\{x\})^T \mid y\}\} \tag{2.119}$$

and next develop the inner expectation. That expectation can be written as

$$\begin{aligned}\mathscr{E}\{&[(x - \mathscr{E}\{x \mid y\}) + (\mathscr{E}\{x \mid y\} - \mathscr{E}\{x\})] \\ &\cdot [(x - \mathscr{E}\{x \mid y\}) + (\mathscr{E}\{x \mid y\} - \mathscr{E}\{x\})]^T \mid y\}.\end{aligned} \tag{2.120}$$

Now $\mathscr{E}\{x \mid y\} - \mathscr{E}\{x\}$ is a function of y. Using (2.111), we therefore compute the cross terms in (2.120) to be zero. As a result, (2.120) reduces to

$$\mathscr{E}\{(x - \mathscr{E}\{x \mid y\})(x - \mathscr{E}\{x \mid y\})^T \mid y\} + (\mathscr{E}\{x \mid y\} - \mathscr{E}\{x\})(\mathscr{E}\{x \mid y\} - \mathscr{E}\{x\})^T, \tag{2.121}$$

where, on the last term, we used (2.114). Taking the expectation of (2.121), we have the desired result (2.117). ▲

Example 2.18. The *conditional characteristic function* of the random vector x given the random vector y is defined by

$$\varphi_{x|y}(u \mid y) = \mathscr{E}\{\exp(iu^T x) \mid y\} \tag{2.122}$$

[compare with (2.83)]. Conditional moments can be obtained from the conditional characteristic function in the same way moments are obtained from the unconditional characteristic function. In view of (2.110),

$$\varphi_x(u) = \mathscr{E}\{\varphi_{x|y}(u \mid y)\}. \quad ▲ \tag{2.123}$$

Example 2.19. The joint Gaussian density of rv's x_1 and x_2 is given in (2.74). The exponent in (2.74) can be rewritten as

$$-\frac{1}{2}\left(\frac{x_1 - \mathscr{E}\{x_1\}}{\sigma\{x_1\}}\right)^2 - \frac{1}{2}\left[\frac{x_2 - \mathscr{E}\{x_2\} - \rho(\sigma\{x_2\}/\sigma\{x_1\})(x_1 - \mathscr{E}\{x_1\})}{(1 - \rho^2)^{1/2}\,\sigma\{x_2\}}\right]^2.$$

Therefore, we can write the joint Gaussian density as

$$p_{x_1,x_2}(x_1, x_2) = [(2\pi)^{1/2}\sigma\{x_1\}]^{-1} \exp\left[-\frac{1}{2}\left(\frac{x_1 - \mathscr{E}\{x_1\}}{\sigma\{x_1\}}\right)^2\right]$$
$$\cdot [(2\pi)^{1/2}\sigma\{x_2\}(1 - \rho^2)^{1/2}]^{-1}$$
$$\cdot \exp\left[-\frac{1}{2}\left(\frac{x_2 - \mathscr{E}\{x_2\} - \rho(\sigma\{x_2\}/\sigma\{x_1\})(x_1 - \mathscr{E}\{x_1\})}{\sigma\{x_2\}(1 - \rho^2)^{1/2}}\right)^2\right]. \quad (2.124)$$

But the first factor in the foregoing is the *Gaussian* density of x_1 [compare with (2.40)]. The second factor is also a Gaussian density with mean

$$\mathscr{E}\{x_2\} + \rho[\sigma\{x_2\}/\sigma\{x_1\}](x_1 - \mathscr{E}\{x_1\}) \quad (2.125)$$

and variance

$$\sigma^2\{x_2\}(1 - \rho^2). \quad (2.126)$$

Comparing (2.124) with (2.100), we conclude that the second factor in (2.124) is the conditional density

$$p_{x_2|x_1}(x_2 \mid x_1)$$

and that this density is Gaussian with parameters given in (2.125) and (2.126). Therefore, we have that, if x_1 and x_2 are jointly normally distributed, then the conditional distribution of x_2 given x_1 is also normal with

$$\mathscr{E}\{x_2 \mid x_1\} = \mathscr{E}\{x_2\} + \operatorname{cov}\{x_2, x_1\} \operatorname{var}^{-1}\{x_1\}(x_1 - \mathscr{E}\{x_1\}), \quad (2.127)$$

$$\mathscr{E}\{(x_2 - \mathscr{E}\{x_2 \mid x_1\})^2 \mid x_1\} = \operatorname{var}\{x_2\} - \operatorname{cov}\{x_2, x_1\} \operatorname{var}^{-1}\{x_1\} \operatorname{cov}\{x_1, x_2\}. \quad (2.128)$$

In loose notation, we sometimes say that the "random variable" $x_2 \mid x_1$ is Gaussian. ▲

6. PROPERTIES OF GAUSSIAN RANDOM VARIABLES

The density function of a scalar Gaussian rv was given in (2.40) and its characteristic function in (2.44). It was noted that the density function is characterized by two parameters, the mean and variance of the random variable. It was also noted that the Gaussian density function is symmetric, and, as a result, all the odd central moments are zero. All the even central moments were expressed in terms of the variance in Example 2.6. If the rv's x_1 and x_2 are jointly normal, then, as noted below (2.74), x_1 and x_2 are marginally normal. Furthermore, as shown in Example 2.19, the conditional rv $x_2 \mid x_1$ is also normal. According to Theorem 2.6,

if x_1 and x_2 are uncorrelated, then they are independent. These properties of Gaussian random variables, their generalizations, and additional properties to be developed here, account for their importance in mathematics. Their engineering importance will become apparent in subsequent chapters.

Let the random $(n + m)$-vector $z^T = [x^T, y^T]$, where x is an n-vector and y is an m-vector, be $N \sim (m_z, P_z)$. That is, its density and characteristic functions [see (2.80) and (2.84)] are given by

$$p_z(z) = [(2\pi)^{(n+m)/2} \mid P_z \mid^{1/2}]^{-1} \exp[-\tfrac{1}{2}(z - m_z)^T P_z^{-1}(z - m_z)], \quad (2.129)$$

$$\varphi_z(u) = \exp(iu^T m_z - \tfrac{1}{2}u^T P_z u), \quad (2.130)$$

where

$$m_z^T = \mathscr{E}\{z^T\} = [\mathscr{E}\{x^T\}, \quad \mathscr{E}\{y^T\}] = [m_x^T, \quad m_y^T], \quad (2.131)$$

$$\begin{aligned} P_z &= \mathscr{E}\{(z - m_z)(z - m_z)^T\} \\ &= \begin{bmatrix} \mathscr{E}\{(x - m_x)(x - m_x)^T\}, & \mathscr{E}\{(x - m_x)(y - m_y)^T\} \\ \mathscr{E}\{(y - m_y)(x - m_x)^T\}, & \mathscr{E}\{(y - m_y)(y - m_y)^T\} \end{bmatrix} \\ &= \begin{bmatrix} P_x & P_{xy} \\ P_{yx} & P_y \end{bmatrix}, \end{aligned} \quad (2.132)$$

$$\mid P_z \mid = \mid P_x \mid \mid P_y - P_{yx}P_x^{-1}P_{xy} \mid = \mid P_y \mid \mid P_x - P_{xy}P_y^{-1}P_{yx} \mid .^{11} \quad (2.133)$$

Now assume that the elements of x are uncorrelated with the elements of y. That is, $P_{xy} = P_{yx}^T = 0$. Then we can write the joint characteristic function in (2.130) as

$$\begin{aligned} \varphi_z(u) &= \varphi_{x,y}(u_1, u_2) \\ &= \exp\left[i[u_1^T, u_2^T] \begin{bmatrix} m_x \\ m_y \end{bmatrix} - \tfrac{1}{2}[u_1^T, u_2^T] \begin{bmatrix} P_x & 0 \\ 0 & P_y \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}\right] \\ &= \exp(iu_1^T m_x - \tfrac{1}{2}u_1^T P_x u_1) \exp(iu_2^T m_y - \tfrac{1}{2}u_2^T P_y u_2). \end{aligned} \quad (2.134)$$

But these two exponential functions are characteristic functions of normal random vectors, one $N \sim (m_x, P_x)$, the other $N \sim (m_y, P_y)$. In view of (2.131) and (2.132), these are the characteristic functions of x and y, respectively. Therefore, we have proved an extension to Theorem 2.6.

[11] See Bellman [2, p. 103]. P_z is assumed positive definite here.

Theorem 2.10. *If the Gaussian random vectors* x *and* y *are uncorrelated, they are independent. Furthermore,* x *and* y *are marginally Gaussian (with appropriate parameters).*

Theorem 2.11. *Let the p-vector* $\mathrm{z} \sim N(m_z, P_z)$. *Let* $\mathrm{w} = C\mathrm{z} + a$, *where C is a* $q \times p$ *constant matrix, a is a constant q-vector, and* w *is a random q-vector. Then* $\mathrm{w} \sim N(Cm_z + a, CP_zC^T)$.

PROOF: Write the characteristic function of w. By definition,

$$\varphi_w(u) = \mathscr{E}\{\exp(iu^T\mathrm{w})\} = \exp(iu^Ta)\,\mathscr{E}\{\exp(iu^TC\mathrm{z})\}$$
$$= \exp(iu^Ta)\,\varphi_z(C^Tu).$$

But z is Gaussian, so that

$$\varphi_z(C^Tu) = \exp(iu^TCm_z - \tfrac{1}{2}u^TCP_zC^Tu).$$

Therefore,

$$\varphi_w(u) = \exp[iu^T(Cm_z + a) - \tfrac{1}{2}u^TCP_zC^Tu].$$

But this is the characteristic function of a Gaussian random vector with mean $Cm_z + a$ and covariance matrix CP_zC^T. ■

The meaning of Theorem 2.11 is essentially that linear operations on Gaussian random vectors produce Gaussian random vectors.

Theorem 2.12. *If the random vectors* x *and* y *are jointly normally distributed [according to* (2.129)], *then* y (*say*) *is also normally distributed (with appropriate parameters).*

PROOF: Let $\mathrm{w}^T = [\mathrm{w}_1^T, \mathrm{w}_2^T]$, and consider the transformation

$$\mathrm{w}_1 = \mathrm{x} - P_{xy}P_y^{-1}\mathrm{y}, \qquad \mathrm{w}_2 = \mathrm{y}. \tag{2.135}$$

In view of Theorem 2.11, w is normally distributed. An easy computation shows that w_1 and w_2 are uncorrelated. In fact,

$$\mathscr{E}\{(\mathrm{w}_1 - \mathscr{E}\{\mathrm{w}_1\})(\mathrm{w}_1 - \mathscr{E}\{\mathrm{w}_1\})^T\}$$
$$= \mathscr{E}\{[(\mathrm{x} - m_x) - P_{xy}P_y^{-1}(\mathrm{y} - m_y)][(\mathrm{x} - m_x) - P_{xy}P_y^{-1}(\mathrm{y} - m_y)]^T\}$$
$$= P_x - P_{xy}P_y^{-1}P_{yx},$$

so that

$$\mathscr{E}\{(\mathrm{w} - \mathscr{E}\{\mathrm{w}\})(\mathrm{w} - \mathscr{E}\{\mathrm{w}\})^T\} = \begin{bmatrix} P_x - P_{xy}P_y^{-1}P_{yx}, & 0 \\ 0, & P_y \end{bmatrix}. \tag{2.136}$$

Therefore, by Theorem 2.10, w_1 and w_2 are normal. But $w_2 = y$, and so y is normal. ■

Theorem 2.13. *Let* x *and* y *be jointly normally distributed* [*according to* (2.129)]. *Then the conditional density of* x *given* y *is normal with mean*

$$m_x + P_{xy}P_y^{-1}(y - m_y) \tag{2.137}$$

and covariance matrix

$$P_x - P_{xy}P_y^{-1}P_{yx}\,. \tag{2.138}$$

(*That is,* x | y *is normal with these parameters.*)

PROOF: We have seen in the proof of Theorem 2.12 that w_1 and w_2 in (2.135) are normal and independent. As a result, their joint density (also normal) is the product of their individual densities

$$w_1 \sim N(m_x - P_{xy}P_y^{-1}m_y\,,\ P_x - P_{xy}P_y^{-1}P_{yx}), \tag{2.139}$$

$$w_2 \sim N(m_y\,,\ P_y), \tag{2.140}$$

as is evident from (2.135) and (2.136). That is,

$$\begin{aligned} p_{w_1,w_2}(w_1\,,\,w_2) = {} & [(2\pi)^{n/2} \mid P_x - P_{xy}P_y^{-1}P_{yx} \mid^{1/2}]^{-1} \\ & \cdot \exp\{-\tfrac{1}{2}(w_1 - m_x + P_{xy}P_y^{-1}m_y)^T (P_x - P_{xy}P_y^{-1}P_{yx})^{-1} \\ & \cdot (w_1 - m_x + P_{xy}P_y^{-1}m_y)\} \\ & \cdot [(2\pi)^{m/2} \mid P_y \mid^{1/2}]^{-1} \exp\{-\tfrac{1}{2}(w_2 - m_y)^T P_y^{-1}(w_2 - m_y)\}. \end{aligned} \tag{2.141}$$

Now the Jacobian of the transformation (2.135) is 1, and so, according to Theorem 2.7, we obtain the density $p_{x,y}(x, y)$ by eliminating w_1 and w_2, in (2.141), in favor of x and y via (2.135). As a result, we get

$$\begin{aligned} p_{x,y}(x, y) = {} & [(2\pi)^{n/2} \mid P_x - P_{xy}P_y^{-1}P_{yx} \mid^{1/2}]^{-1} \\ & \cdot \exp\{-\tfrac{1}{2}[x - m_x - P_{xy}P_y^{-1}(y - m_y)]^T [P_x - P_{xy}P_y^{-1}P_{yx}]^{-1} \\ & \cdot [x - m_x - P_{xy}P_y^{-1}(y - m_y)]\} \\ & \cdot [(2\pi)^{m/2} \mid P_y \mid^{1/2}]^{-1} \exp\{-\tfrac{1}{2}(y - m_y)^T P_y^{-1}(y - m_y)\}. \end{aligned} \tag{2.142}$$

[Note that, in view of (2.133), the coefficient in (2.142) agrees with that in (2.129). In fact, we have just indirectly *proved* (2.133) !] Now the second density in (2.142) is the marginal density of y (Theorem 2.12). Therefore, a comparison with (2.100) (definition of conditional density function) reveals that the first density is $p_{x|y}(x \mid y)$. ■

In the proofs of Theorems 2.12 and 2.13, it was assumed that the covariance matrix P_z is positive definite. It can be shown that (2.137) and (2.138) are valid even if P_z is singular. See Anderson [1, Chapter 2] for the singular case.

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